TPS2HB35-Q1 40-V, 35-mΩ Dual-Channel Smart High-Side Switch

1 Features

- Qualified for automotive applications
- AEC-Q100 qualified with the following results:
  - Device temperature grade 1: T_A = –40°C to 125°C ambient operating temperature range
  - Device HBM ESD classification level 2
  - Device CDM ESD classification level C4B
  - Withstands 40-V load dump
- Dual-channel smart high-side switch with 35-mΩ R_{ON} (T_J = 25°C)
- Improve system level reliability through adjustable current limiting
  - Current limit adjustable from 2 A to 25 A
- Robust integrated output protection:
  - Integrated thermal protection
  - Protection against short to ground/battery
  - Protection against reverse battery events including automatic switch on with reverse voltage
  - Automatic shut off if loss of battery/ground occurs
  - Integrated output clamp to demagnetize inductive loads
  - Configurable fault handling
- Analog sense output can be configured to accurately measure:
  - Load current
  - Device temperature
- Provides fault indication through SNS pin
  - Detection of open load and short-to-battery

2 Applications

- Automotive display module
- ADAS modules
- Seat comfort module
- Transmission control unit
- HVAC control module
- Body control modules
- LED lighting

3 Description

The TPS2HB35-Q1 device is a dual-channel smart high-side switch intended for use in 12-V automotive systems. The device integrates robust protection and diagnostic features to ensure output port protection even during harmful events like short circuits in automotive systems. The device protects against faults through a reliable current limit, which, depending on device variant, is adjustable from 2 A to 25 A .

The high current limit range allows for usage in loads that require large transient currents, while the low current limit range provides improved protection for loads that do not require high peak current. The device is capable of reliably driving various load profiles.

The TPS2HB35-Q1 also provides a high accuracy analog current sense that allows for improved load diagnostics. By reporting load current and device temperature to a system MCU, the device enables predictive maintenance and load diagnostics that improves the system lifetime.

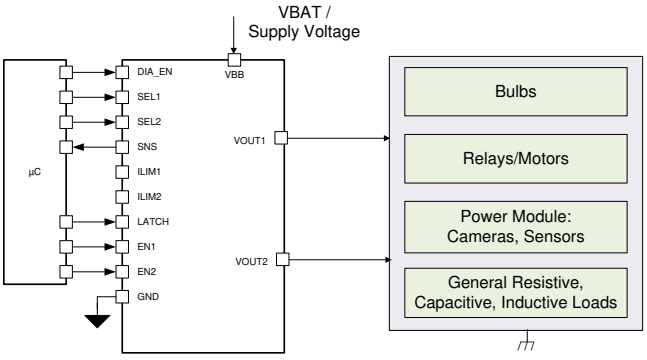
The TPS2HB35-Q1 is available in a HTSSOP package which allows for reduced PCB footprint.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
TPS2HB35-Q1	HTSSOP (16)	5.00 mm × 4.40 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Simplified Schematic



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

Changes from Revision C (November 2019) to Revision D — **Page**

- Added Versions A and B to the *Electrical Characteristics* table in the *Specifications* section ... 6

Changes from Revision B (July 2019) to Revision C — **Page**

- Changed from Advance Information to Production Data ... 1

Changes from Revision A (April 2019) to Revision B — **Page**

- Added Device Version C to the Current Limit Characteristics in the *Specifications Electrical Characteristics* table ... 6

Changes from Original (February 2018) to Revision A — **Page**

- Changes made throughout the data sheet ... 1

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5 Device Comparison Table

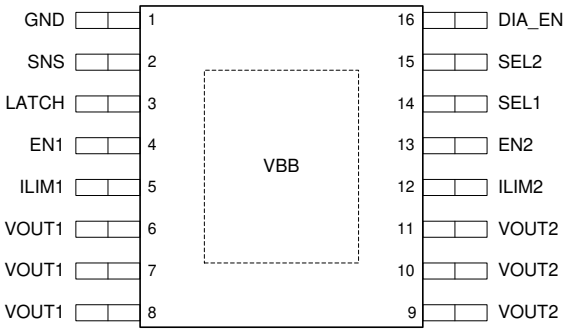
Table 1. TPS2HB35-Q1 Device Options

Device Version	Part Number	Current Limit	Current Limit Range	Overcurrent Behavior
A	TPS2HB35*A*QPWPRQ1	Resistor Programmable	2 A - 10 A	Disable switch immediately
B	TPS2HB35*B*QPWPRQ1	Resistor Programmable	5.6 A - 25 A	Disable switch immediately
C	TPS2HB35*C*QPWPRQ1	Resistor Programmable	2.5 A - 6 A	Switch stays on until thermal shutdown

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6 Pin Configuration and Functions

PWP Package (Version A/B/C)
16-Pin HTSSOP
Top View



Pin Functions

PIN		I/O	DESCRIPTION
NO.	NAME		
1	GND	—	Device ground
2	SNS	O	Sense output
3	LATCH	I	Sets fault handling behavior (latched or auto-retry)
4	EN1	I	Channel 1 control input, active high
5	ILIM1	O	Connect pull-up resistor to V_{BB} to set current-limit threshold on CH1
6-8	VOUT1	O	Channel 1 output
9-11	VOUT2	O	Channel 2 output
12	ILIM2	O	Connect pull-up resistor to V_{BB} to set current-limit threshold on CH2
13	EN2	I	Channel 2 control input, active high
14	SEL1	I	Diagnostics select 1
15	SEL2	I	Diagnostics select 2
16	DIA_EN	I	Diagnostic enable, active high
Exposed pad	VBB	I	Power supply input

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6.1 Recommended Connections for Unused Pins

The TPS2HB35-Q1 device is designed to provide an enhanced set of diagnostic and protection features. However, if the system design only allows for a limited number of I/O connections, some pins may be considered optional.

Table 2. Connections for Optional Pins

PIN NAME	CONNECTION IF NOT USED	IMPACT IF NOT USED
SNS	Ground through 1-kΩ resistor	Analog sense is not available.
LATCH	Float or ground through R_{PROT} resistor	With LATCH unused, the device will auto-retry after a fault. If latched behavior is desired, but the system describes limited I/O, it is possible to use one microcontroller output to control the latch function of several high-side channels.
ILIM1, ILIM2	Float	If the ILIMx pin is left floating, the device will be set to the default internal current-limit threshold.
SEL1	Float or ground through R_{PROT} resistor	SEL1 selects the T_J sensing feature. With SEL1 unused, only CH1 and CH2 current sensing and open load detection are available.
SEL2	Ground through R_{PROT} resistor	With SEL2 = 0 V, CH2 current sensing and CH2 open load detection are not available.
DIA_EN	Float or ground through R_{PROT} resistor	With DIA_EN unused, the analog sense, open-load, and short-to-battery diagnostics are not available.

R_{PROT} is used to protect the pins from excess current flow during reverse battery conditions, for more information see the section on *Reverse Battery* protection.

7 Specifications

7.1 Absolute Maximum Ratings

Over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
Maximum continuous supply voltage, V_{BB}			36	V
Load dump voltage, V_{LD}	ISO16750-2:2010(E)		40	V
Reverse battery voltage, V_{Rev}, t ≤ 3 minutes		–18		V
Enable pin voltage, V_{EN1} and V_{EN2}		–1	7	V
LATCH pin voltage, V_{LATCH}		–1	7	V
Diagnostic Enable pin voltage, V_{DIA_EN}		–1	7	V
Sense pin voltage, V_{SNS}		–1	18	V
Select pin voltage, V_{SEL1} and V_{SEL2}		–1	7	V
Reverse ground current, I_{GND}	V_{BB} < 0 V		–50	mA
Energy dissipation during turnoff, E_{TOFF}	Single pulse, one channel, L_{OUT} = 5 mH, $T_{J,start}$ = 125°C		42(2)	mJ
Energy dissipation during turnoff, E_{TOFF}	Repetitive pulse, one channel, L_{OUT} = 5 mH, $T_{J,start}$ = 125°C		20(2)	mJ
Maximum junction temperature, T_J			150	°C
Storage temperature, T_{stg}		–65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute maximum rated conditions for extended periods may affect device reliability.
(2) For further details, see the section regarding switch-off of an inductive load.

7.2 ESD Ratings

				VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per AEC Q100-002(1)	All pins except VBB and VOUTx	±2000	V
			VBB and VOUTx	±4000	
		Charged-device model (CDM), per AEC Q100-011	All pins	±750	

(1) AEC-Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specifications.

7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted) (1)

		MIN	MAX	UNIT
V_{BB}	Nominal supply voltage (1)	6	18	V
V_{BB}	Extended supply voltage(2)	3	28	V
V_{EN1}, V_{EN2}	Enable voltage	–1	5.5	V
V_{LATCH}	LATCH voltage	–1	5.5	V
V_{DIA_EN}	Diagnostic Enable voltage	–1	5.5	V
V_{SEL1}, V_{SEL2}	Select voltage	–1	5.5	V
V_{SNS}	Sense voltage	–1	7	V
T_A	Operating free-air temperature	–40	125	°C

(1) All operating voltage conditions are measured with respect to device GND
(2) Device will function within extended operating range, however some parametric values might not apply

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7.4 Thermal Information

THERMAL METRIC(1)(2)		TPS2HB35-Q1 PWP (HTSSOP) 16 PINS	UNIT
$R_{\theta JA}$	Junction-to-ambient thermal resistance	32.9	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	30.8	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	9.0	°C/W
ψ_{JT}	Junction-to-top characterization parameter	1.8	°C/W
ψ_{JB}	Junction-to-board characterization parameter	9.2	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	2.0	°C/W

(1) For more information about traditional and new thermal metrics, see the Semiconductor and IC Package Thermal Metrics application report.
(2) The thermal parameters are based on a 4-layer PCB according to the JESD51-5 and JESD51-7 standards.

7.5 Electrical Characteristics

V_{BB} = 6 V to 18 V, T_J = -40°C to 150°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
INPUT VOLTAGE AND CURRENT						
$V_{DSCLAMP}$	V_{DS} clamp voltage		40		46	V
$V_{BBCLAMP}$	V_{BB} clamp voltage		58		76	V
V_{UVLOF}	V_{BB} undervoltage lockout falling	Measured with respect to the GND pin of the device	2.0		3	V
V_{UVLOR}	V_{BB} undervoltage lockout rising	Measured with respect to the GND pin of the device	2.2		3	V
I_{SB}	Standby current (total device leakage including both MOSFET channels)	V_{BB} = 13.5 V, T_J = 25°C V_{ENx} = V_{DIA_EN} = 0 V, V_{OUT} = 0 V			0.5	µA
		V_{BB} = 13.5 V, T_J = 125°C, V_{ENx} = V_{DIA_EN} = 0 V, V_{OUT} = 0 V			4	µA
IL_{NOM}	Continuous load current, per channel	Two channels enabled, T_{AMB} = 70°C		3.5		A
		One channel enabled, T_{AMB} = 70°C		5		A
$I_{OUT(standby)}$	Output leakage current (per channel)	V_{BB} = 13.5 V, T_J = 25°C V_{ENx} = V_{DIA_EN} = 0 V, V_{OUT} = 0 V		0.01	0.5	µA
		V_{BB} = 13.5 V, T_J = 125°C V_{ENx} = V_{DIA_EN} = 0 V, V_{OUT} = 0 V			1.5	µA
I_{DIA}	Current consumption in diagnostic mode	V_{BB} = 13.5 V, I_{SNS} = 0 mA V_{ENx} = 0 V, V_{DIA_EN} = 5 V, V_{OUT} = 0V		3	6	mA
I_Q	Quiescent current	V_{BB} = 13.5 V V_{ENx} = V_{DIA_EN} = 5 V, I_{OUTx} = 0 A		3	6	mA
t_{STBY}	Standby mode delay time	V_{ENx} = V_{DIA_EN} = 0 V to standby	12	17	22	ms
RON CHARACTERISTICS						
R_{ON}	On-resistance (Includes MOSFET and package)	T_J = 25°C, 6 V ≤ V_{BB} ≤ 28 V, I_{OUT1} = I_{OUT2} > 1 A		35		mΩ
		T_J = 150°C, 6 V ≤ V_{BB} ≤ 28 V, I_{OUT1} = I_{OUT2} > 1 A			75	mΩ
		T_J = 25°C, 3 V ≤ V_{BB} ≤ 6 V, I_{OUT1} = I_{OUT2} > 1 A			70	mΩ
$R_{ON(REV)}$	On-resistance during reverse polarity	T_J = 25°C, -18 V ≤ V_{BB} ≤ -8 V		45		mΩ
		T_J = 105°C, -18 V ≤ V_{BB} ≤ -8 V			70	mΩ
CURRENT SENSE CHARACTERISTICS						
K_{SNS}	Current sense ratio I_{OUTx} / I_{SNS}	I_{OUTX} = 1 A		2000		

Electrical Characteristics (continued)

V_{BB} = 6 V to 18 V, T_J = -40°C to 150°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
I_{SNSI}	Current sense current and accuracy	V_{EN} = V_{DIA_EN} = 5 V, V_{SEL1} = 0 V, V_{SEL2} = X	I_{OUT} = 3 A		1.5		mA
				–5		5	%
			I_{OUT} = 1 A		0.5		mA
				–5		5	%
			I_{OUT} = 300 mA		0.150		mA
				-6		6	%
			I_{OUT} = 100 mA		0.05		mA
				-7		7	%
			I_{OUT} = 50 mA		0.025		mA
				-13		13	%
I_{SNSI}	Current sense current and accuracy	V_{EN} = V_{DIA_EN} = 5 V, V_{SEL1} = 0 V, V_{SEL2} = X	I_{OUT} = 20 mA		0.00894		mA
I_{SNSI}	Current sense current and accuracy	V_{EN} = V_{DIA_EN} = 5 V, V_{SEL1} = 0 V, V_{SEL2} = X		-35		35	%
TJ SENSE CHARACTERISTICS							
I_{SNST}	Temperature sense current Device Version A/B/C	V_{DIA_EN} = 5 V, V_{SEL1} = 5 V, V_{SEL2} = 0 V	T_J = -40°C	0.00	0.12	0.29	mA
			T_J = 25°C	0.68	0.85	1.02	mA
			T_J = 85°C	1.25	1.52	1.79	mA
			T_J = 125°C	1.61	1.96	2.31	mA
			T_J = 150°C	1.80	2.25	2.70	mA
dI_{SNST}/dT	Coefficient				0.011		mA/°C
SNS CHARACTERISTICS							
I_{SNSFH}	I_{SNS} fault high-level	V_{DIA_EN} = 5 V, V_{SEL1} = 0 V, V_{SEL2} = X		4	4.5	5.3	mA
$I_{SNSleak}$	I_{SNS} leakage	V_{DIA_EN} = 0 V				1	µA
CURRENT LIMIT CHARACTERISTICS							
$I_{CL,max}$	Current Limit Max	Device Version C, T_J = -40°C to 150°C	R_{ILIM} = 8.25 kΩ		13		A
			R_{ILIM} = 10 kΩ		12.5		A
			R_{ILIM} = 15 kΩ		11.5		A
			R_{ILIM} = 25 kΩ		9		A
I_{CL}	Current Limit Threshold	Device Version C, T_J = -40°C to 150°C	R_{ILIM} = GND, open, or out of range		14		A
			R_{ILIM} = 8.25 kΩ	4.4	6	8.4	A
			R_{ILIM} = 25 kΩ	1.52	2.5	3.48	A
		Device Version A, T_J = -40°C to 150°C	R_{ILIM} = GND, open, or out of range		14		A
			R_{ILIM} = 5 kΩ	7.8	10	12.5	A
			R_{ILIM} = 25 kΩ	1.8	2	2.5	A
		Device Version B, T_J = -40°C to 150°C	R_{ILIM} = GND, open, or out of range		36		A
			R_{ILIM} = 5 kΩ	18.5	25	30	A
			R_{ILIM} = 25 kΩ	4.6	5.6	7.5	A
K_{CL}	Current Limit Ratio	Version A/C			50		A * kΩ
		Version B			140		A * kΩ
FAULT CHARACTERISTICS							
V_{OL}	Open-load (OL) detection voltage	V_{ENx} = 0 V, V_{DIA_EN} = 5 V		2	3	4	V

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Electrical Characteristics (continued)

V_{BB} = 6 V to 18 V, T_J = -40°C to 150°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
t_{OL1}	OL and STB indication-time from ENx falling	V_{ENx} = 5 V to 0 V, V_{DIA_EN} = 5 V, V_{SEL1} = 0 V(1) I_{OUT} = 0 mA, V_{OUTx} = 4 V		300	500	700	µs
t_{OL2}	OL and STB indication-time from DIA_EN rising	V_{ENx} = 0 V, V_{DIA_EN} = 0 V to 5 V, V_{SEL1} = 0 V(1) I_{OUT} = 0 mA, V_{OUTx} = 4 V				50	µs
t_{OL3}	OL and STB indication-time from V_{OUT} rising	V_{ENx} = 0 V, V_{DIA_EN} = 5 V, V_{SEL1} = 0 V(1) I_{OUT} = 0 mA, V_{OUTx} = 0 V to 4 V				50	µs
T_{ABS}	Thermal shutdown			150			°C
T_{REL}	Relative thermal shutdown	For Version A/B only			60		°C
T_{REL}	Relative thermal shutdown	For Version C only	For Version C only		80		°C
T_{HYS}	Thermal shutdown hysteresis				28		°C
t_{FAULT}	Fault shutdown indication-time	V_{DIA_EN} = 5 V Time between switch shutdown and I_{SNS} settling at I_{SNSFH}				50	µs
t_{RETRY}	Retry time	Time from fault shutdown until switch re-enable (thermal shutdown or current limit).		1	2	3	ms
EN1 AND EN2 PIN CHARACTERISTICS(2)							
$V_{IL, ENx}$	Input voltage low-level	No GND network diode				0.8	V
$V_{IH, ENx}$	Input voltage high-level	No GND network diode		2			V
$V_{IHYS, ENx}$	Input voltage hysteresis				350		mV
R_{ENx}	Internal pulldown resistor			0.5	1	2	MΩ
$I_{IL, EN}$	Input current low-level	V_{EN} = 0.8 V			0.8		µA
$I_{IH, EN}$	Input current high-level	V_{EN} = 5 V			5		µA
DIA_EN PIN CHARACTERISTICS(2)							
V_{IL, DIA_EN}	Input voltage low-level	No GND network diode				0.8	V
V_{IH, DIA_EN}	Input voltage high-level	No GND network diode		2.0			V
V_{IHYS, DIA_EN}	Input voltage hysteresis			200	350	530	mV
R_{DIA_EN}	Internal pulldown resistor			0.5	1	2	MΩ
I_{IL, DIA_EN}	Input current low-level	V_{DIA_EN} = 0.8 V			0.8		µA
I_{IH, DIA_EN}	Input current high-level	V_{DIA_EN} = 5 V			5.0		µA
SEL1 AND SEL2 PIN Characteristics							
$V_{IL, SELx}$	Input voltage low-level	No GND network diode				0.8	V
$V_{IH, SELx}$	Input voltage high-level	No GND network diode		2			V
$V_{IHYS, SELx}$	Input voltage hysteresis				350		mV
R_{SELx}	Internal pulldown resistor			0.5	1	2	MΩ
$I_{IL, SELX}$	Input current low-level	V_{SELX} = 0.8 V			0.8		µA
$I_{IH, SELX}$	Input current high-level	V_{SELX} = 5 V			5		µA
LATCH PIN CHARACTERISTICS(2)							
$V_{IL, LATCH}$	Input voltage low-level	No GND network diode				0.8	V
$V_{IH, LATCH}$	Input voltage high-level	No GND network diode		2.0			V
$V_{IHYS, LATCH}$	Input voltage hysteresis			200	350	530	mV
R_{LATCH}	Internal pulldown resistor			0.5	1	2	MΩ
$I_{IL, LATCH}$	Input current low-level	V_{LATCH} = 0.8 V			0.8		µA

(1) SELx must be set to select the relevant channel. Diagnostics are performed on Channel 1 when SELx = 00 and diagnostics are performed on channel 2 when SELx = 01
(2) V_{BB} = 3 V to 28 V

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Electrical Characteristics (continued)

V_{BB} = 6 V to 18 V, T_J = -40°C to 150°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
$I_{IH, LATCH}$	Input current high-level	V_{LATCH} = 5 V		5		µA

7.6 SNS Timing Characteristics

V_{BB} = 6 V to 18 V, T_J = -40°C to +150°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
SNS TIMING - CURRENT SENSE						
$t_{SNSION1}$	Settling time from rising edge of DIA_EN	V_{ENx} = 5 V, V_{DIA_EN} = 0 V to 5 V R_{SNS} = 1 kΩ, R_L ≤ 5 Ω			40	µs
$t_{SNSION2}$	Settling time from rising edge of ENx and DIA_EN	V_{ENx} = V_{DIA_EN} = 0 V to 5 V R_{SNS} = 1 kΩ, R_L ≤ 5 Ω			165	µs
$t_{SNSION3}$	Settling time from rising edge of ENx	V_{ENx} = 0 V to 5 V, V_{DIA_EN} = 5 V R_{SNS} = 1 kΩ, R_L ≤ 5 Ω			165	µs
$t_{SNSIOFF1}$	Settling time from falling edge of DIA_EN	V_{ENx} = 5 V, V_{DIA_EN} = 5 V to 0 V R_{SNS} = 1 kΩ, R_L ≤ 5 Ω			20	µs
$t_{SETTLEH}$	Settling time from rising edge of load step	V_{EN1} = 5 V, V_{DIA_EN} = 5 V R_{SNS} = 1 kΩ, I_{OUT} = 5 A to 1 A			20	µs
$t_{SETTLEL}$	Settling time from falling edge of load step	V_{ENx} = 5 V, V_{DIA_EN} = 5 V R_{SNS} = 1 kΩ, I_{OUT} = 5 A to 1 A			20	µs
SNS TIMING - TEMPERATURE SENSE						
$t_{SNSTON1}$	Settling time from rising edge of DIA_EN	V_{ENx} = 5 V, V_{DIA_EN} = 0 V to 5 V R_{SNS} = 1 kΩ			40	µs
$t_{SNSTON2}$	Settling time from rising edge of DIA_EN	V_{ENx} = 0 V, V_{DIA_EN} = 0 V to 5 V R_{SNS} = 1 kΩ			70	µs
$t_{SNSTOFF}$	Settling time from falling edge of DIA_EN	V_{ENx} = X, V_{DIA_EN} = 5 V to 0 V R_{SNS} = 1 kΩ			20	µs
SNS TIMING - MULTIPLEXER						
t_{MUX}	Settling time from temperature sense to current sense	V_{ENx} = X, V_{DIA_EN} = 5 V V_{SEL1} = 5 V to 0 V, V_{SEL2} = X R_{SNS} = 1 kΩ, R_L ≤ 5 Ω			60	µs
	Settling time from current sense on CHx to CHy	V_{ENx} = X, V_{DIA_EN} = 5 V V_{SEL1} = 0 V, V_{SEL2} = 0 V to 5 V R_{SNS} = 1 kΩ, I_{OUT1} = 2 A, I_{OUT2} = 4 A			20	µs
	Settling time from current sense to temperature sense	V_{ENx} = X, V_{DIA_EN} = 5 V V_{SEL1} = 0 V to 5 V, V_{SEL2} = X R_{SNS} = 1 kΩ, R_L ≤ 5 Ω			60	µs

7.7 Switching Characteristics

V_{BB} = 13.5 V, T_J = -40°C to +150°C (unless otherwise noted)

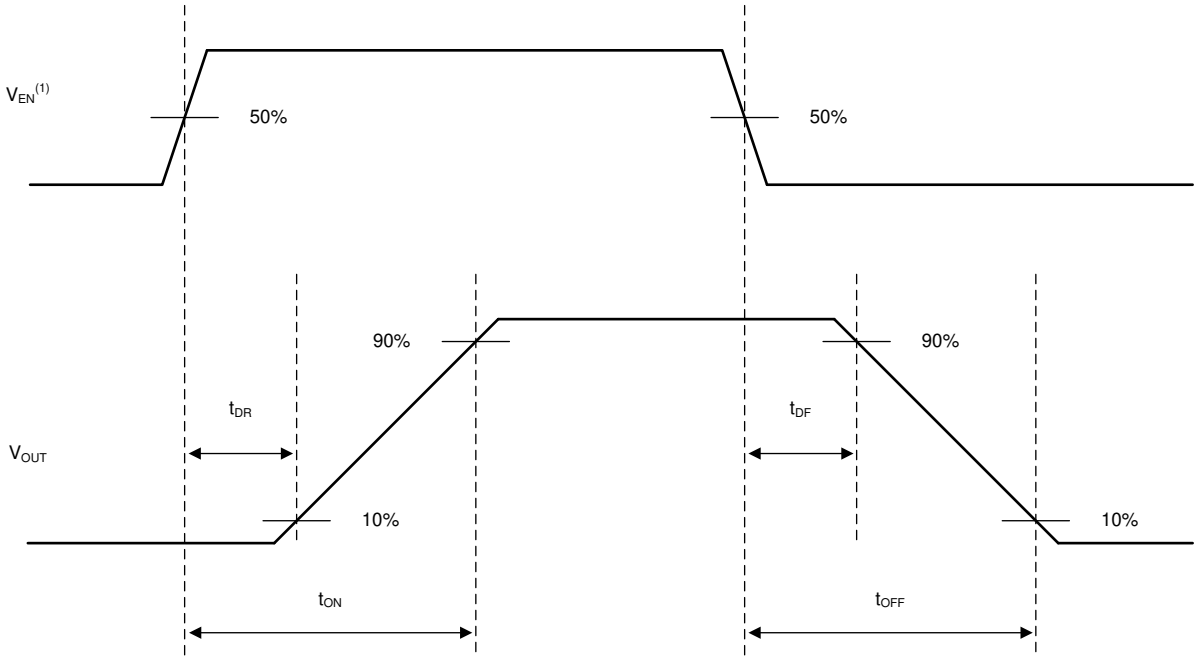
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{DR}	Turnon delay time	V_{BB} = 13.5 V, R_L ≤ 5 Ω, 50% EN rising to 10% V_{OUT} rising	20	60	100	µs
t_{DF}	Turnoff delay time	V_{BB} = 13.5 V, R_L ≤ 5 Ω, 50% EN falling to 90% V_{OUT} Falling	20	60	100	µs
SR_R	VOUTx rising slew rate	V_{BB} = 13.5 V, 20% to 80% of V_{OUT}, R_L ≤ 5 Ω	0.1	0.4	0.7	V/µs
SR_F	VOUTx falling slew rate	V_{BB} = 13.5 V, 80% to 20% of V_{OUT}, R_L ≤ 5 Ω	0.1	0.4	0.7	V/µs
t_{ON}	Turnon time	V_{BB} = 13.5 V, R_L ≤ 5 Ω, 50% EN rising to 80% V_{OUT} rising	39	87	145	µs
t_{OFF}	Turnoff time	V_{BB} = 13.5 V, R_L ≤ 5 Ω, 50% EN rising to 80% V_{OUT} rising	39	87	147	µs

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Switching Characteristics (continued)

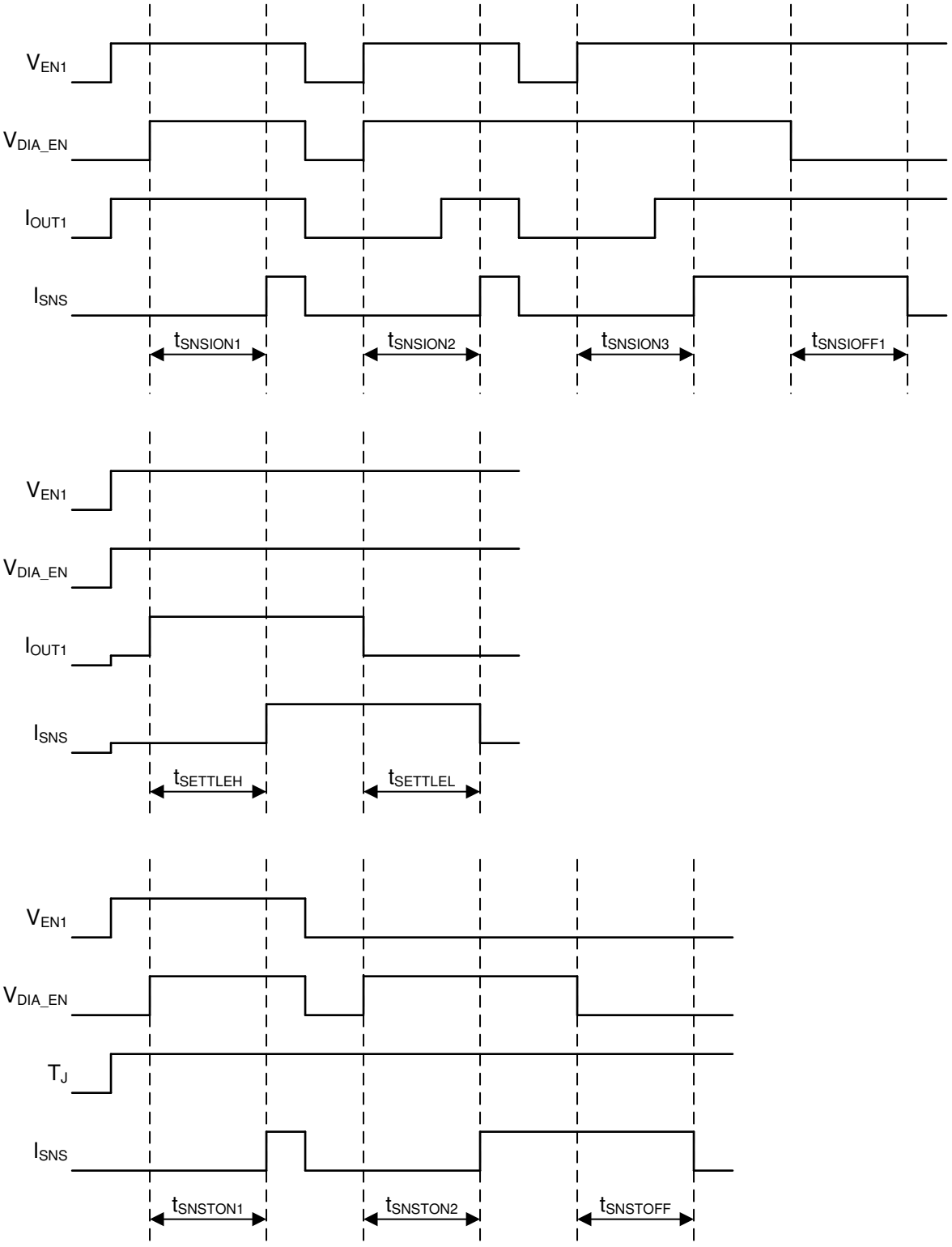
V_{BB} = 13.5 V, T_J = -40°C to +150°C (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{ON} - t_{OFF}	Turnon and turnoff matching	200-µs enable pulse	–50	0	50	µs
E_{ON}	Switching energy losses during turnon	V_{BB} = 13.5 V, R_L ≤ 5 Ω		0.6		mJ
E_{OFF}	Switching energy losses during turnoff	V_{BB} = 13.5 V, R_L ≤ 5 Ω		0.6		mJ



Rise and fall time of V_{ENx} is 100 ns.

Figure 1. Switching Characteristics Definitions



NOTE1: Rise and fall times of control signals are 100 ns. Control signals include: EN1, EN2, DIA_EN, SEL1, SEL2.

NOTE2: SEL1 and SEL2 must be set to the appropriate values.

Figure 2. SNS Timing Characteristics Definitions

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7.8 Typical Characteristics

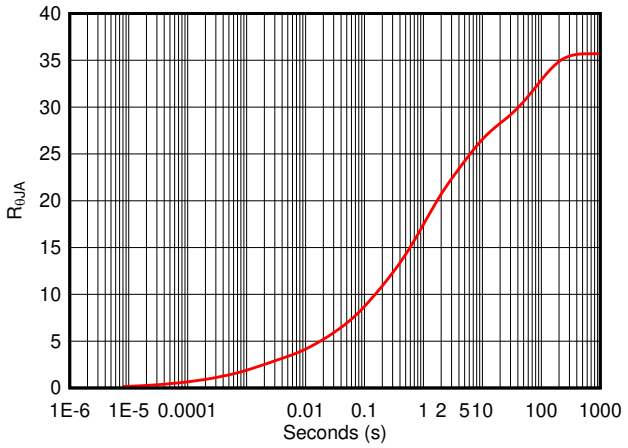


Figure 3. Transient Thermal Impedance 1 Channel Enabled

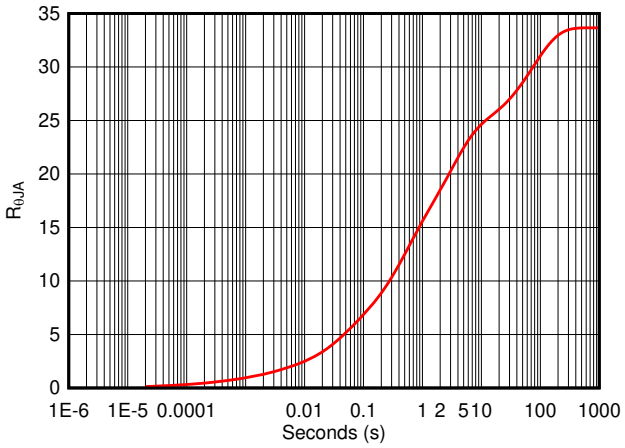


Figure 4. Transient Thermal Impedance Both Channels Enabled

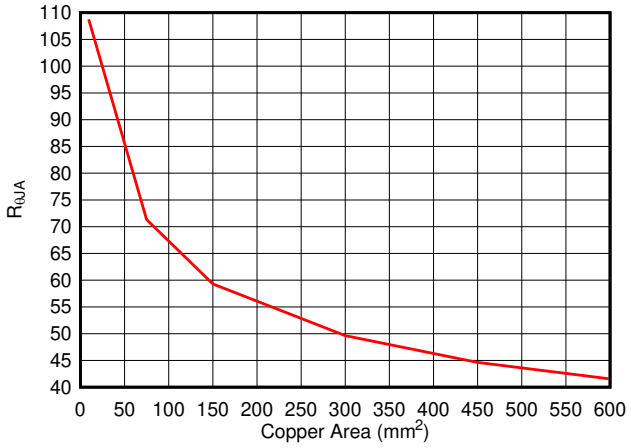


Figure 5. $R_{\theta JA}$ vs Copper Area

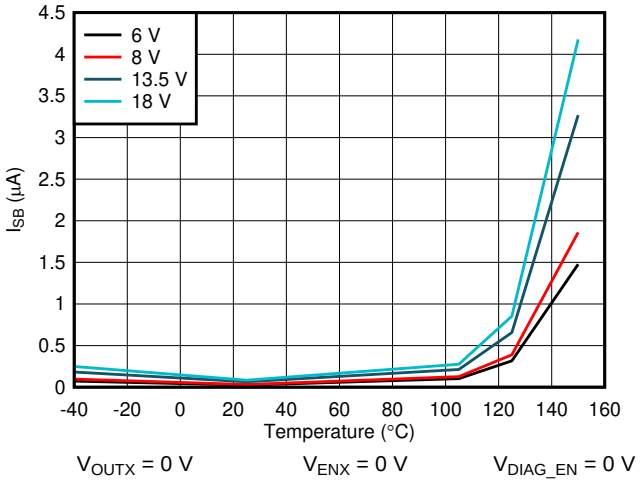
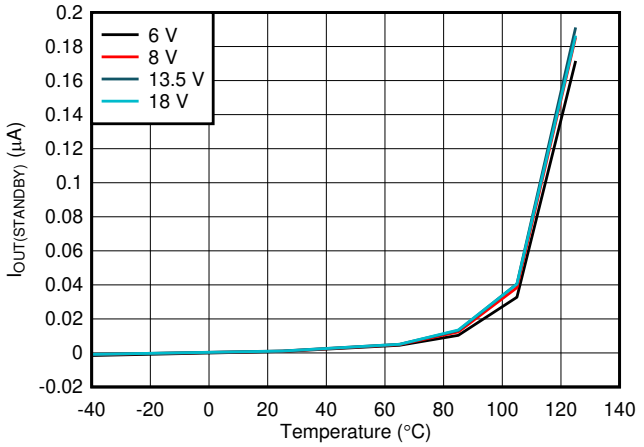


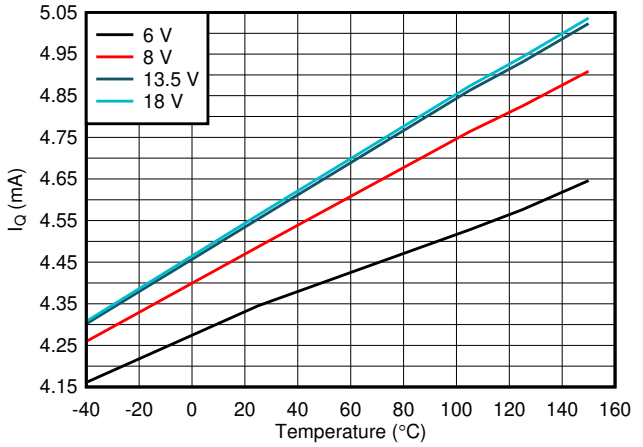
Figure 6. Standby Current (I_{SB}) vs Temperature

Typical Characteristics (continued)



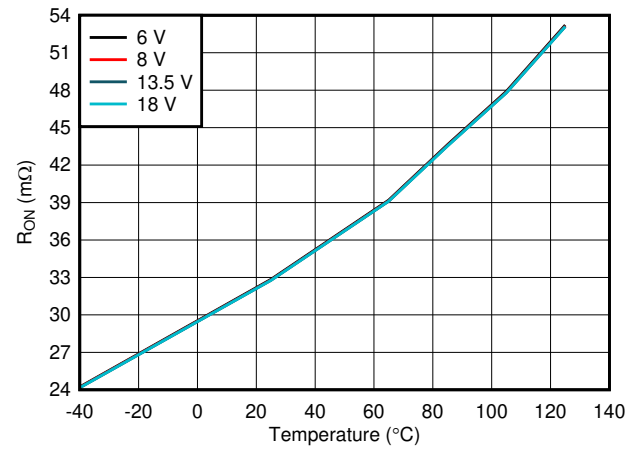
V_{OUTX} = 0 V V_{ENX} = 0 V V_{DIAG_EN} = 0 V
Both Channels

Figure 7. Output Leakage Current ($I_{OUT(standby)}$) vs Temperature



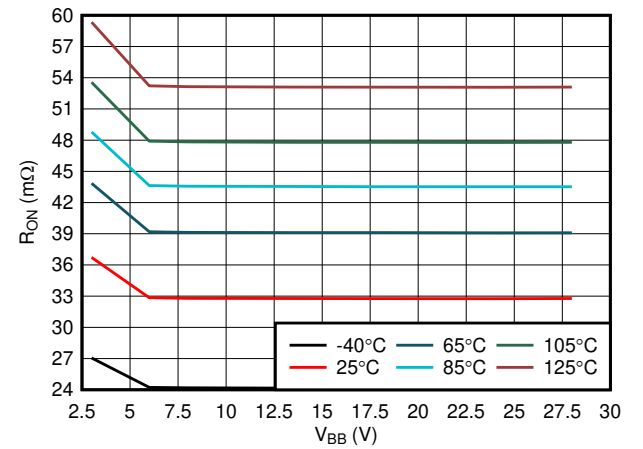
I_{OUTX} = 0 A V_{ENX} = 5 V V_{DIAG_EN} = 5 V
R_{SNS} = 1 kΩ V_{SEL1} = V_{SEL2} = 0 V

Figure 8. Quiescent Current (I_Q) vs Temperature



I_{OUTX} = 200 mA V_{ENX} = 5 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ

Figure 9. On Resistance (R_{ON}) vs Temperature

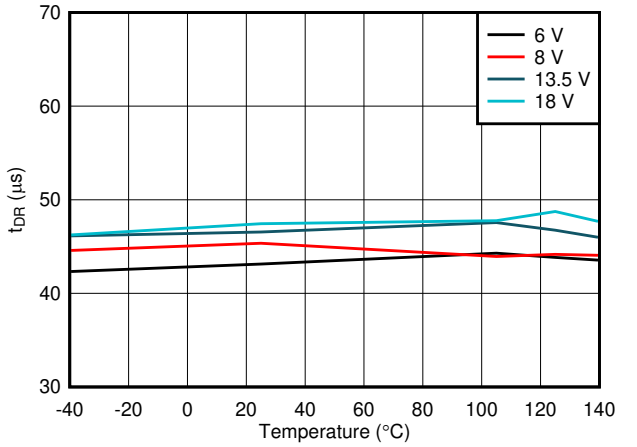


I_{OUTX} = 200 mA V_{ENX} = 5 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ

Figure 10. On Resistance (R_{ON}) vs V_{BB}

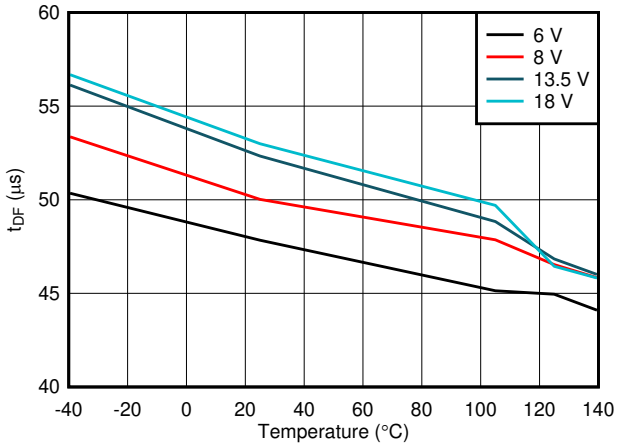
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Typical Characteristics (continued)



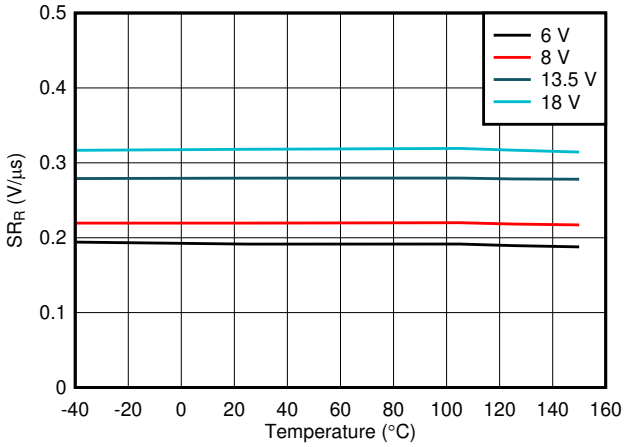
R_{OUTX} = 2.6 Ω V_{ENX} = 0 V to 5 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V Both Channels

Figure 11. Turn-on Delay Time (t_{DR}) vs Temperature



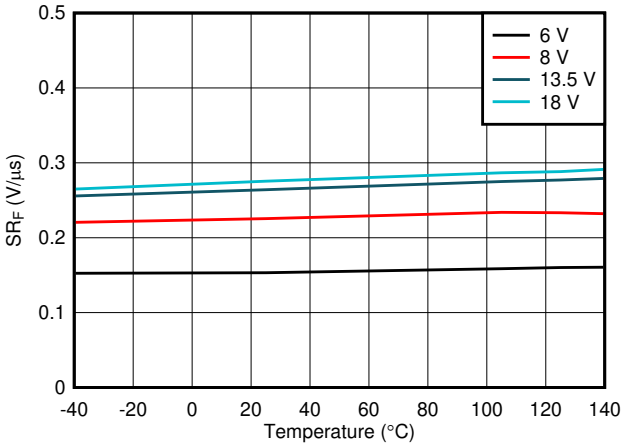
R_{OUTX} = 2.6 Ω V_{ENX} = 5 V to 0 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V Both Channels

Figure 12. Turn-off Delay Time (t_{DF}) vs Temperature



R_{OUTX} = 2.6 Ω V_{ENX} = 0 V to 5 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V Both Channels

Figure 13. V_{OUT} Slew Rate Rising (SR_R) vs Temperature

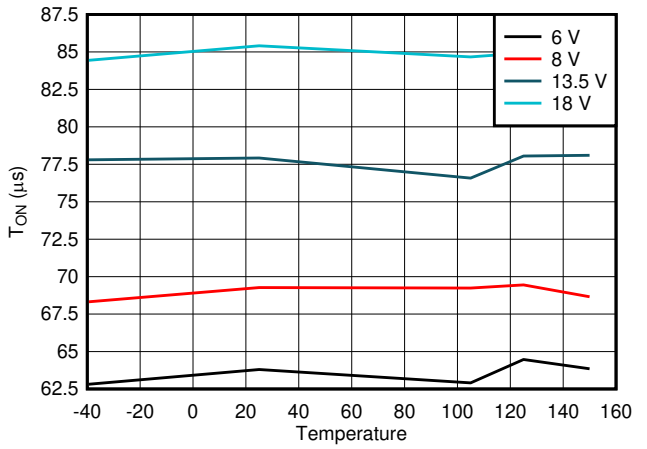


R_{OUTX} = 2.6 Ω V_{ENX} = 5 V to 0 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V Both Channels

Figure 14. V_{OUT} Slew Rate Falling (SR_F) vs Temperature

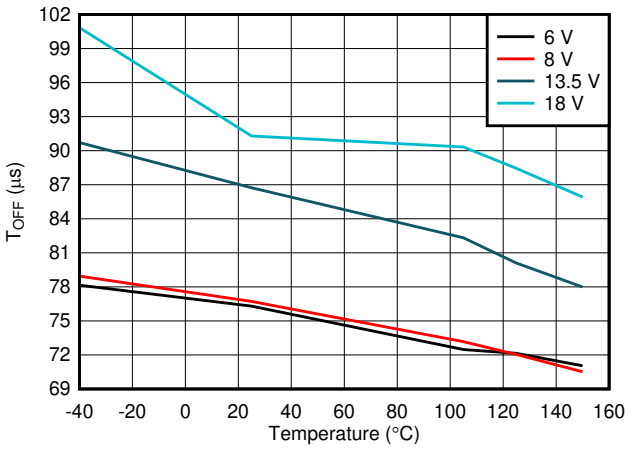
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Typical Characteristics (continued)



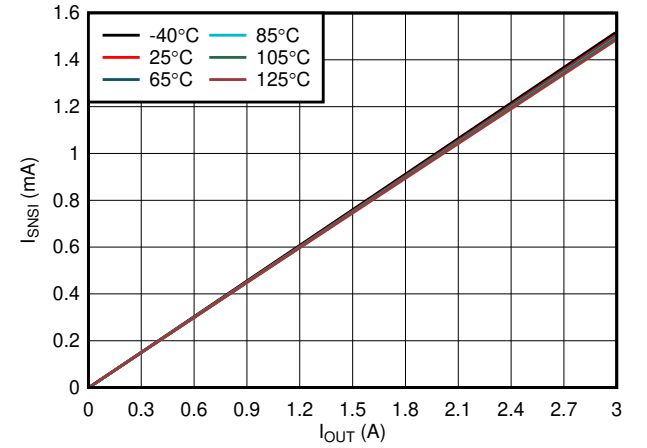
R_{OUTX} = 2.6 Ω V_{ENX} = 0 V to 5 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V Both Channels

Figure 15. Turn-on Time (t_{ON}) vs Temperature



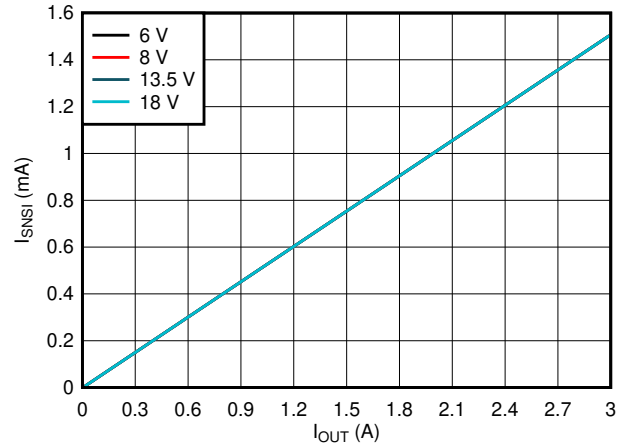
R_{OUTX} = 2.6 Ω V_{ENX} = 5 V to 0 V V_{DIAG_EN} = 0 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V Both Channels

Figure 16. Turn-off Time (t_{OFF}) vs Temperature



V_{SEL1} = V_{SEL2} = 0 V V_{ENX} = 5 V V_{DIAG_EN} = 5 V
R_{SNS} = 1 kΩ V_{BB} = 13.5 V

Figure 17. Current Sense Output Current (I_{SNSI}) vs Load Current (I_{OUT}) Across Temperature

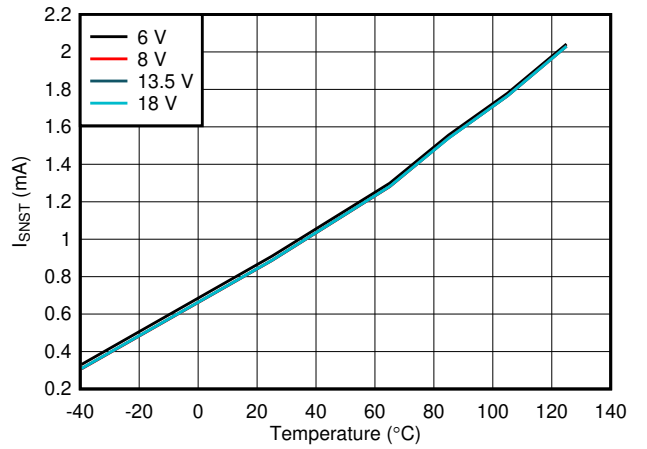


V_{SEL1} = V_{SEL2} = 0 V V_{ENX} = 5 V V_{DIAG_EN} = 5 V
R_{SNS} = 1 kΩ T_A = 25°C

Figure 18. Current Sense Output Current (I_{SNSI}) vs Load Current (I_{OUT}) Across V_{BB}

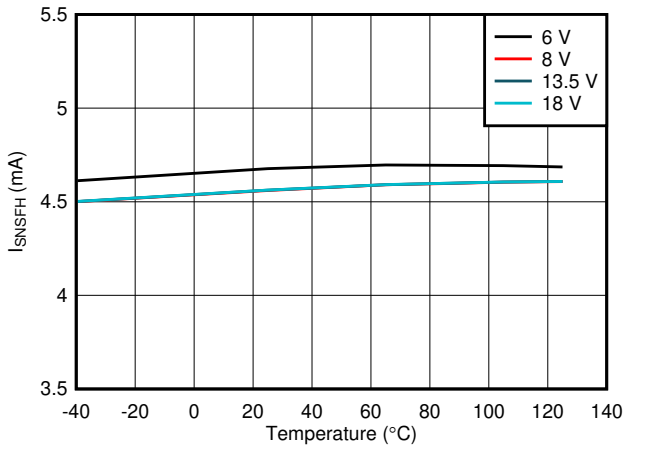
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Typical Characteristics (continued)



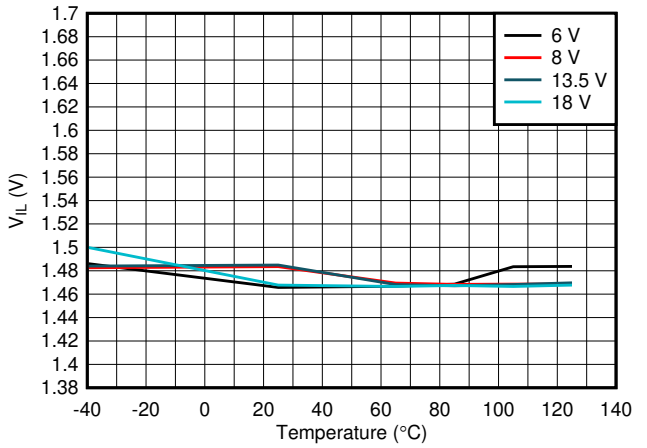
V_{SEL1} = 5 V V_{SEL2} = 0 V V_{DIAG_EN} = 5 V
R_{SNS} = 1 kΩ V_{ENX} = 0 V

Figure 19. Temperature Sense Output Current (I_{SNST}) vs Temperature



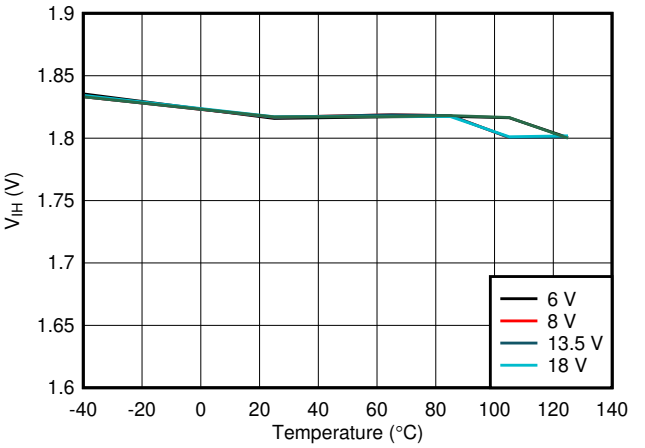
V_{SEL1} = V_{SEL2} = 0 V V_{ENX} = 0 V V_{DIAG_EN} = 5 V
R_{SNS} = 500 Ω V_{OUTX} Floating Both Channels

Figure 20. Fault High Output Current (I_{SNSFH}) vs Temperature



V_{ENX} = 3.3 V to 0 V V_{OUTX} = 0 V V_{DIAG_EN} = 0 V
R_{OUTX} = 1 kΩ

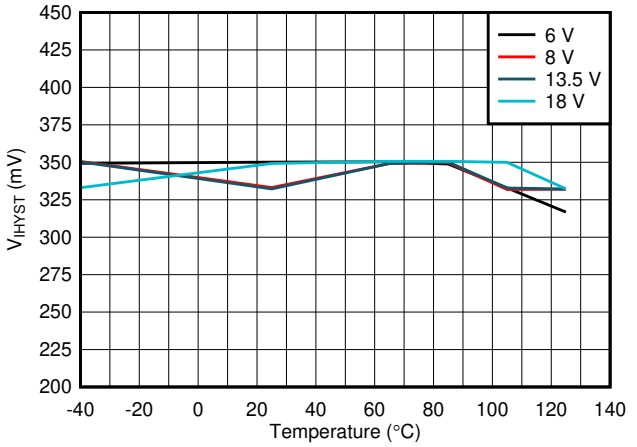
Figure 21. V_{IL} vs Temperature



V_{ENX} = 0 V to 3.3 V V_{OUTX} = 0 V V_{DIAG_EN} = 0 V
R_{OUTX} = 1 kΩ

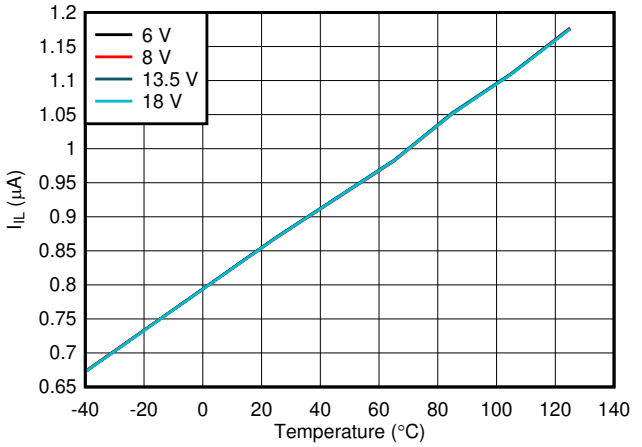
Figure 22. V_{IH} vs Temperature

Typical Characteristics (continued)



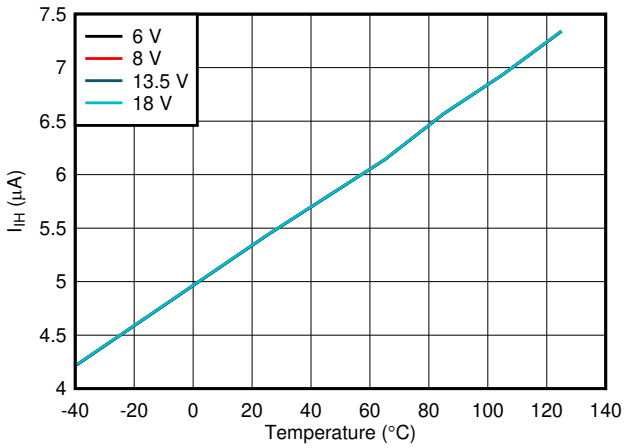
V_{ENX} = 0 V to 3.3 V and 3.3 V to 0 V
R_{OUTX} = 1 kΩ
V_{OUTX} = 0 V
V_{DIAG_EN} = 0 V

Figure 23. V_{IHYS} vs Temperature



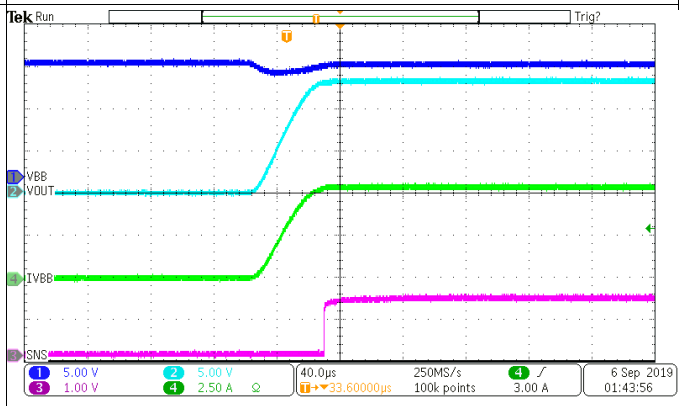
V_{ENX} = 0.8 V
R_{OUTX} = 1 kΩ
V_{OUTX} = 0 V
V_{DIAG_EN} = 0 V

Figure 24. I_{IL} vs Temperature



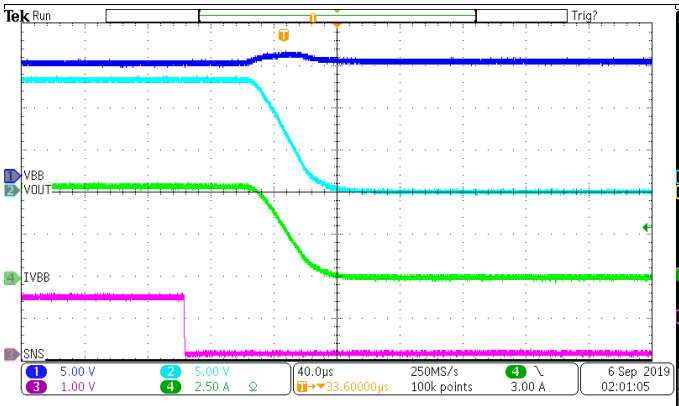
V_{ENX} = 5 V
R_{OUTX} = 1 kΩ
V_{OUTX} = 0 V
V_{DIAG_EN} = 0 V

Figure 25. I_{IH} vs Temperature



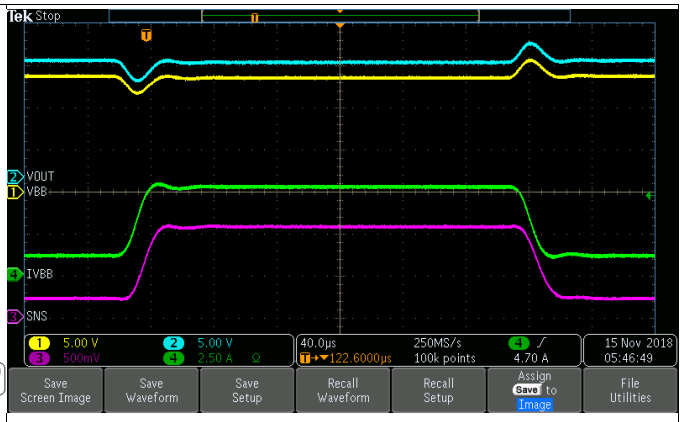
R_{OUT1} = 5 Ω
V_{SEL1} = V_{SEL2} = 0 V
R_{SNS} = 1 kΩ
V_{DIA_EN} = 5 V

Figure 26. Turn-on Time (t_{ON})



R_{OUT1} = 5 Ω
V_{SEL1} = V_{SEL2} = 0 V
R_{SNS} = 1 kΩ
V_{DIA_EN} = 5 V

Figure 27. Turn-off Time (t_{OFF})

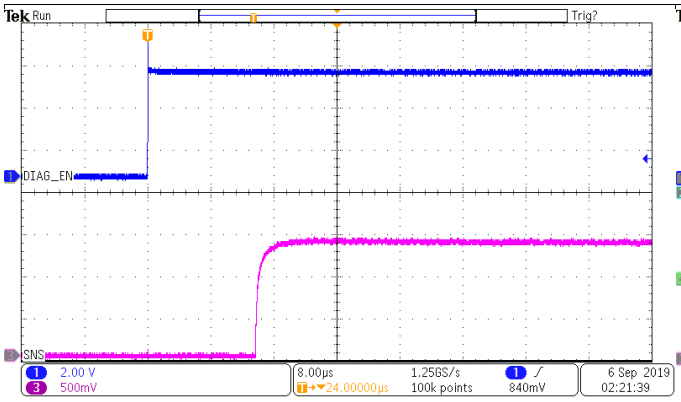


R_{OUT1} = 2.6 Ω
I_{OUT1} = 1 A to 5 A
R_{SNS} = 1 kΩ
V_{BB} = 13.5 V
V_{SEL1} = V_{SEL2} = 0 V

Figure 28. I_{SNS} Settling time ($t_{SNSION1}$) on Load Step

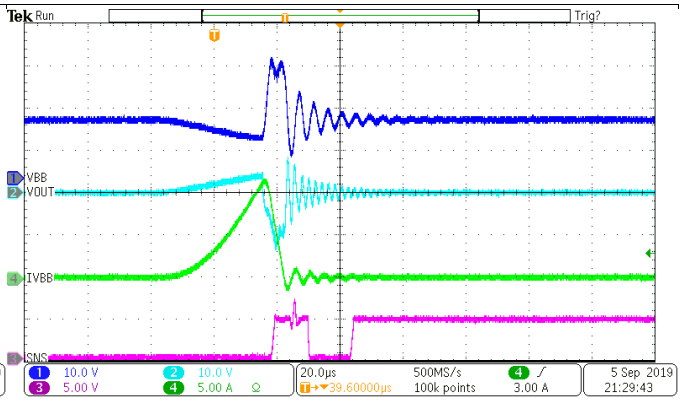
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Typical Characteristics (continued)



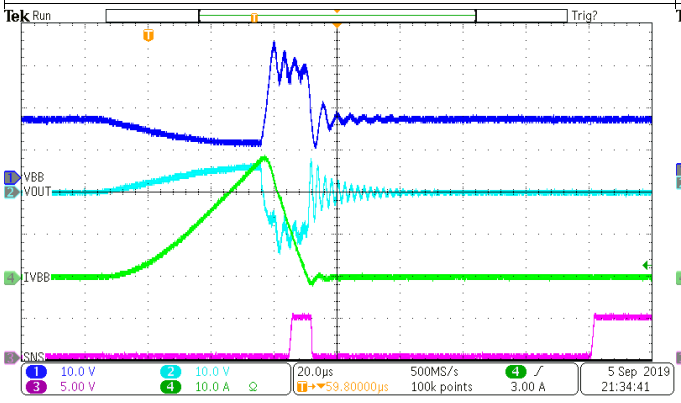
V_{BB} = 13.5 V
V_{EN} = 0 V to 5 V
T_A = 25°C
R_{OUT1} = 5 Ω

Figure 29. SNS Output Current Measurement Enable on DIAG_EN PWM



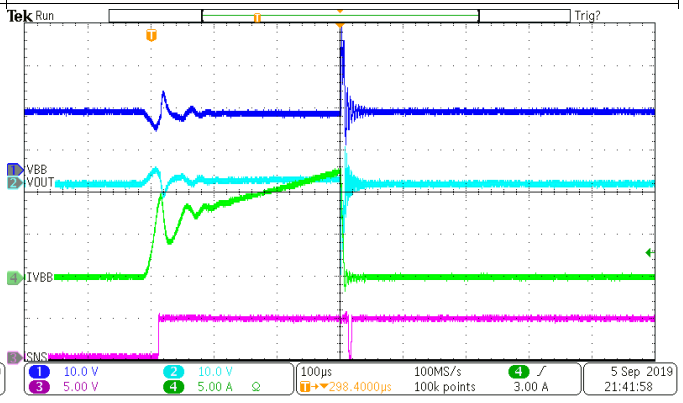
L_{OUT} = 5 µH to GND
V_{EN} = 0 V to 5 V
R_{SNS} = 1 kΩ
V_{DIAG_EN} = 5 V
V_{SEL1} = V_{SEL2} = 0 V
T_A = 25°C

Figure 30. Device Version A Short Circuit Event



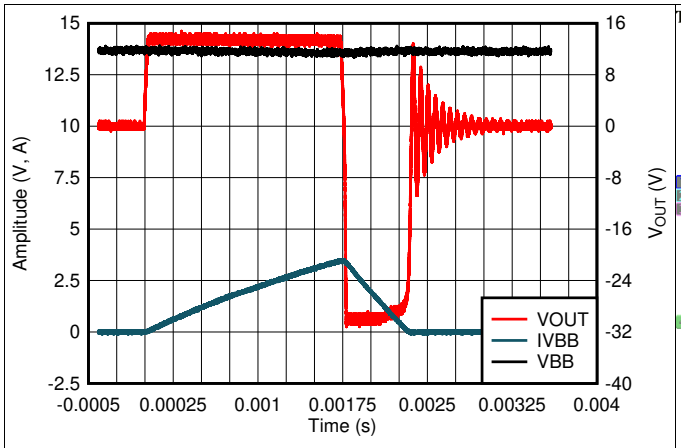
L_{OUT} = 5 µH to GND
V_{EN} = 0 V to 5 V
R_{SNS} = 1 kΩ
V_{DIAG_EN} = 5 V
V_{SEL1} = V_{SEL2} = 0 V
T_A = 25°C

Figure 31. Device Version B Short Circuit Event



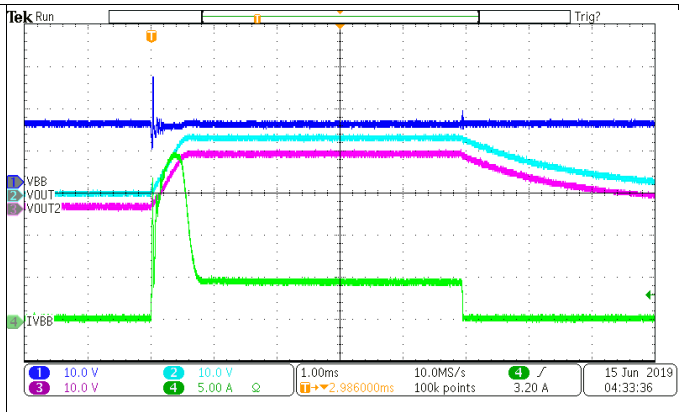
L_{OUT} = 5 µH to GND
V_{EN} = 0 V to 5 V
R_{SNS} = 1 kΩ
V_{DIAG_EN} = 5 V
V_{SEL1} = V_{SEL2} = 0 V
T_A = 25°C

Figure 32. Device Version C Short Circuit Event



V_{BB} = 13.5 V
T_A = 25°C
L_{OUT} = 5 mH

Figure 33. 5 mH Inductive Load Demagnetization



V_{BB} = 13.5 V
R^{OUT} = 6 Ω
T_A = 25°C
C_{OUT} = 270 µF

Figure 34. TPS2HB35C-Q1 Charging a 270uF Capacitor

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8 Parameter Measurement Information

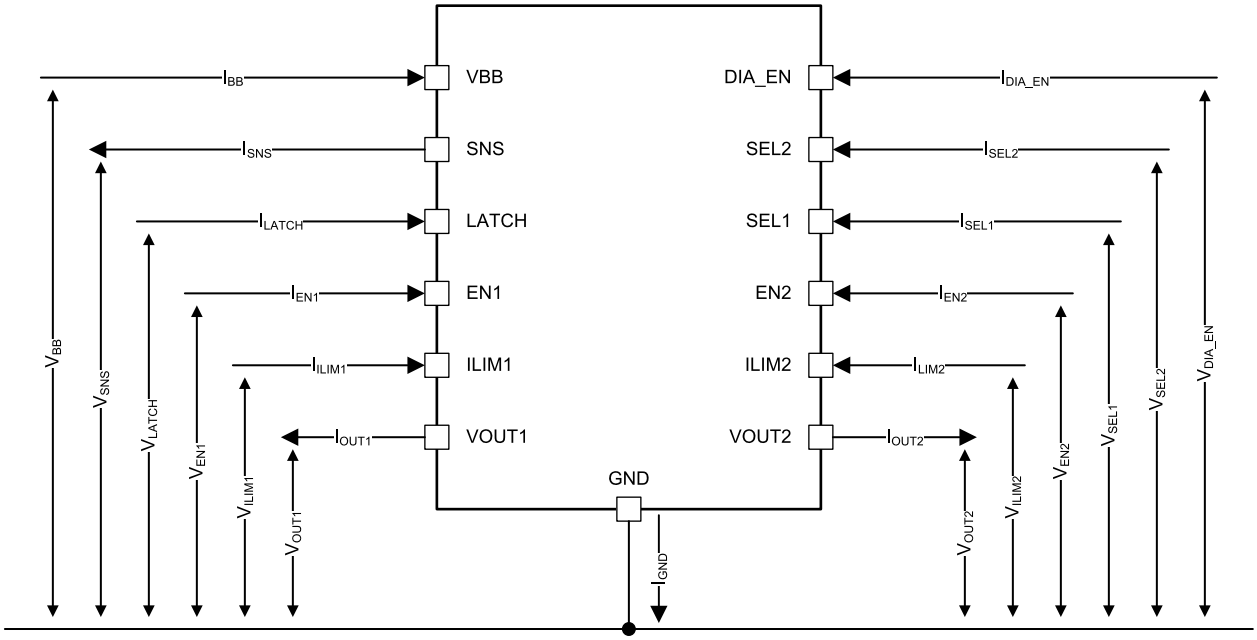


Figure 35. Parameter Definitions

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9 Detailed Description

9.1 Overview

The TPS2HB35-Q1 device is a dual-channel smart high-side switch intended for use with 12-V automotive batteries. Many protection and diagnostic features are integrated in the device.

Diagnostics features include the analog SNS output that is capable of providing a signal that is proportional to load current or device temperature. The high-accuracy load current sense allows for diagnostics of complex loads.

This device includes protection through thermal shutdown, current limiting, transient withstand, and reverse battery operation. For more details on the protection features, refer to the *Feature Description* and *Application Information* sections of the document.

The TPS2HB35-Q1 is one device in a family of TI high side switches. For each device, the part number indicates elements of the device behavior. Figure 36 gives an example of the device nomenclature.

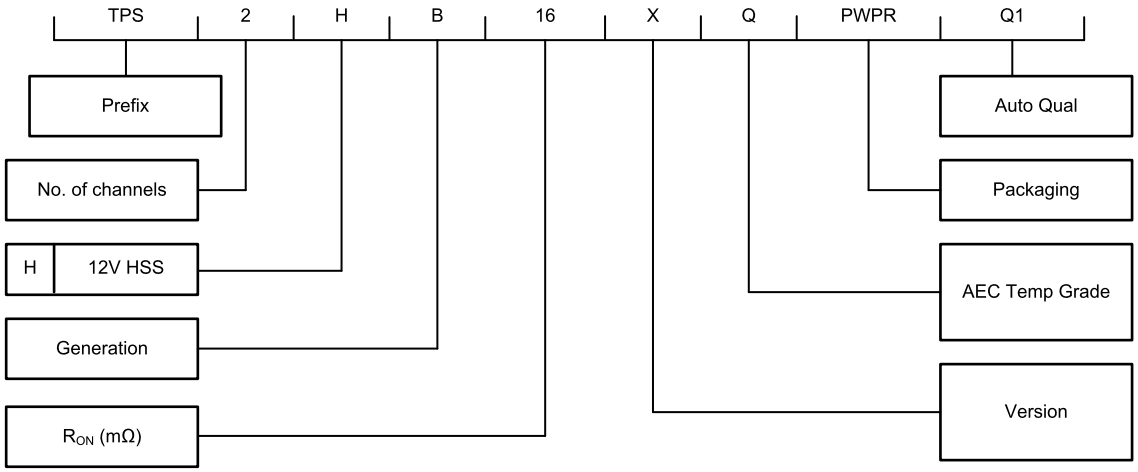
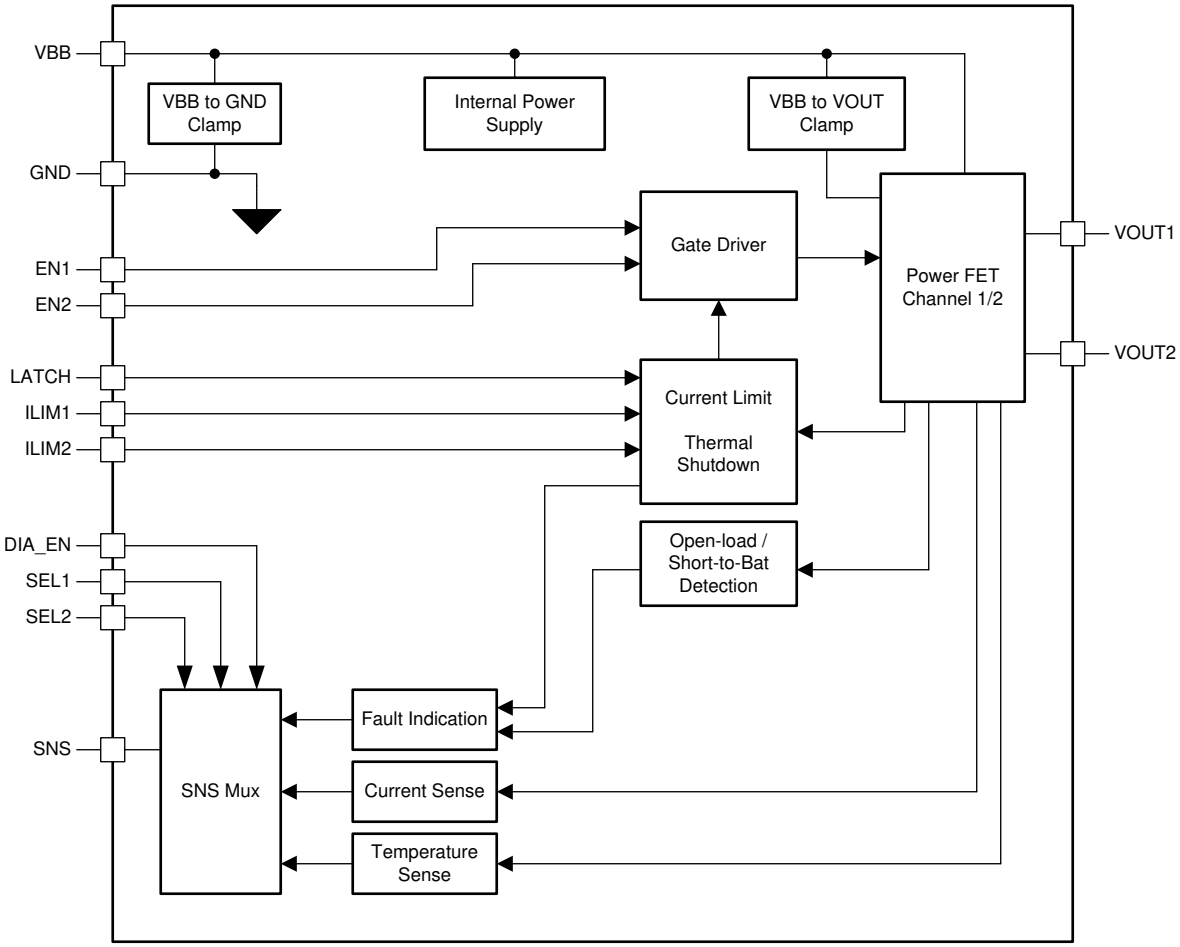


Figure 36. Naming Convention

9.2 Functional Block Diagram

The functional block diagram shown is for device versions A/B/C.



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9.3 Feature Description

9.3.1 Protection Mechanisms

The TPS2HB35-Q1 is designed to operate in the automotive environment. The protection mechanisms allow the device to be robust against many system-level events such as load dump, reverse battery, short-to-ground, and more.

There are two protection features which, if triggered, will cause the switch to automatically disable:

- Thermal Shutdown
- Current Limit

When any of these protections are triggered, the device will enter the FAULT state. In the FAULT state, the fault indication will be available on the SNS pin (see the *Diagnostic Mechanisms* section of the data sheet for more details).

The switch is no longer held off and the fault indication is reset when all of the below conditions are met:

- LATCH pin is low
- t_{RETRY} has expired
- All faults are cleared (thermal shutdown, current limit)

NOTE

CH1 and CH2 operate independently. If there is a fault on one channel, the other channel is not affected.

9.3.1.1 Thermal Shutdown

The device includes a temperature sensor on each power FET and also within the controller portion of the device. There are two cases that the device will consider to be a thermal shutdown fault:

- $T_{J,FET} > T_{ABS}$
- $(T_{J,FET} – T_{J,controller}) > T_{REL}$

After the fault is detected, the relevant switch will turn off. Each channel is turned off based on the measurement of temperature sensor for that channel. Therefore, if the thermal fault is detected on only one channel, the other channel continues operation. If $T_{J,FET}$ passes T_{ABS}, the fault is cleared when the switch temperature decreases by the hysteresis value, T_{HYS}. If instead the T_{REL} threshold is exceeded, the fault is cleared after T_{RETRY} passes.

9.3.1.2 Current Limit

When I_{OUT} reaches the current limit threshold, I_{CL}, the channel will switch off immediately. The I_{CL} value will vary with slew rate and a fast current increase that occurs during a powered-on short circuit can temporarily go above the specified I_{CL} value. In the case that the device remains enabled (and limits I_{OUT}), the thermal shutdown protection feature may be triggered due to the high amount of power dissipation in the device. When the switch is in the FAULT state it will output an output current I_{SNSFH} on the SNS pin. In addition, fault indication will occur when the switch is actively limiting current (applicable to version C).

During a short circuit event, the device will hit the I_{CL} value that is listed in the *Electrical Characteristics* table (for the given device version and R_{ILIM}) and then turn the output off or regulate the output current to protect the device. The device will register a short circuit event when the output current exceeds I_{CL}, however the measured maximum current may exceed the I_{CL} value due to the TPS2HB35-Q1 deglitch filter and turn-off time. The device is specified to protect itself during a short circuit event over the nominal supple voltage range (as defined in the *Electrical Characteristics* table) at 125°C.

The current limit specification in the datasheet is based on the part being enabled into a short circuit condition with 5-µH inductor on the input and output and the input resistance being less than 10 mΩ and the output impedance less than 100 mΩ. When the part is enabled into this short circuit condition, the current will rise up to the threshold specified in the *Electrical Characteristics* table before it begins to shut off the current. The deglitch filter time for the device to react to the current threshold is 3 µs. Therefore if you take Version A/B and subtract 3 µs from the maximum current value, the current limit threshold will align with the value specified in the *Electrical Characteristics* table.

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Feature Description (continued)

The current threshold is defined for version C is different than version A or B. For version C, the current through the device continues to flow until the device hits relative thermal shutdown (T_{REL}). For different V_{BB}'s the slope of the current will change. Therefore the intersection point of where each of the slopes for the different V_{BB} values is determined as the current threshold, I_{LIM}, as shown in Figure 37. This behavior allows for the TPS2HB35**C**-Q1 to be able to charge up a 270 µF capacitor without shutting off due to hitting the current limit as versions A and B would.

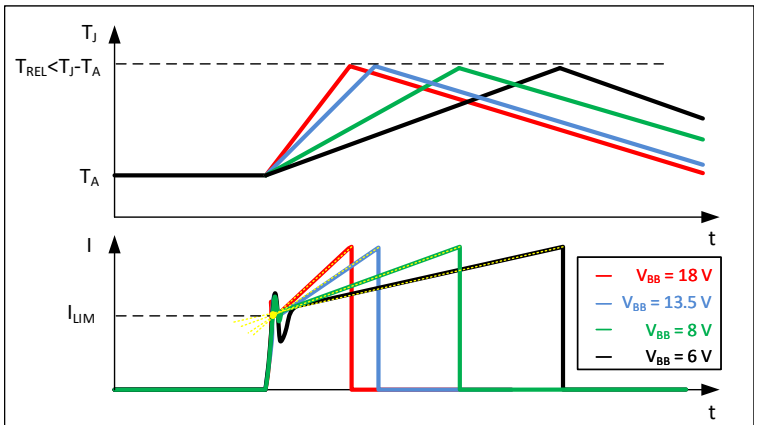


Figure 37. Version C Current Threshold Definition

The maximum point that the current can reach to is bound in the *Electrical Characteristics* table. This worst case value is based on an extremely low impedance short and the current at which T_{REL} is finally reached.

9.3.1.2.1 Current Limit Foldback

Version B of the TPS2HB35-Q1 implement a current limit foldback feature that is designed to protect the device in the case of a long-term fault condition. If the device undergoes fault shutdown events (either of thermal shutdown or current limit) seven consecutive times, the current limit will be reduced to half of the original value. The device will revert back to the original current limit threshold if either of the following occurs:

- The device goes to standby mode.
- The switch turns on and turns off without any fault occurring.

Version A and C do not implement the current limit foldback due to the lower current limit causing less harm during repetitive long-term faults.

9.3.1.2.2 Programmable Current Limit

The TPS2HB35-Q1 includes an adjustable current limit. Some applications (for example, incandescent bulbs) will require a high current limit. Other applications can benefit from a lower current limit threshold. In general, wherever possible a lower current limit is recommended due to allowing system advantages through:

- Reduced size and cost in current carrying components such as PCB traces and module connectors
- Less disturbance at the power supply (V_{BB} pin) during a short circuit event
- Improved protection of the downstream load

To set the current limit threshold, connect a resistor from I_{LIM} to V_{BB}. The current limit threshold is determined by Equation 1 (R_{ILIM} in kΩ):

$$I_{CL} = K_{CL} / R_{ILIM} \quad (1)$$

The R_{ILIM} range is between 5 kΩ and 25 kΩ. An R_{ILIM} resistor is required; however, in the fault case where the pin is floating, grounded, or outside of this range the current limit will default to an internal level that is defined in the *Specifications* section of this document.

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Feature Description (continued)

NOTE

Capacitance on the I_{LIM} pin can cause I_{LIM} to go out of range during short circuit events. For accurate current limiting, place R_{ILIM} near to the device with short traces to ensure <5 pF capacitance to GND on the I_{LIM} pin.

9.3.1.2.3 Undervoltage Lockout (UVLO)

The device monitors the supply voltage V_{BB} to prevent unpredicted behaviors in the event that the supply voltage is too low. When the supply voltage falls down to V_{UVLOF}, the output stage is shut down automatically. When the supply rises up to V_{UVLOR}, the device turns back on.

During an initial ramp of V_{BB} from 0 V at a ramp rate slower than 1 V/ms, V_{EN} pin will have to be$_{BB}$ held low until V_{BB} is above UVLO threshold (with respect to board ground) and the supply voltage to the device has reliably reached above the UVLO condition. For best operation, ensure that V_{BB} has risen above UVLO before setting the V_{EN} pin to high.

9.3.1.2.4 V_{BB} During Short-to-Ground

When V_{OUT} is shorted to ground, the module power supply (V_{BB}) can have a transient decrease. This is caused by the sudden increase in current flowing through the wiring harness cables. To achieve ideal system behavior, it is recommended that the module maintain V_{BB} > 3 V (above the maximum V_{UVLOF}) during V_{OUT} short-to-ground. This is typically accomplished by placing bulk capacitance on the power supply node.

9.3.1.3 Voltage Transients

The TPS2HB35-Q1 device describes two types of voltage clamps which protect the FET against system-level voltage transients. The two different clamps are shown in Figure 38.

The clamp from V_{BB} to GND is primarily used to protect the controller from positive transients on the supply line (for example, ISO7637-2). The clamp from V_{BB} to V_{OUT} is primarily used to limit the voltage across the FET when switching off an inductive load. If the voltage potential from V_{BB} to GND exceeds the V_{BB} clamp level, the clamp will allow current to flow through the device from V_{BB} to GND (Path 2). If the voltage potential from V_{BB} to V_{OUT} exceeds the clamping voltage, the power FET will allow current to flow from V_{BB} to V_{OUT} (Path 3). Additional capacitance from V_{BB} to GND can increase the reliability of the system during ISO 7637 pulse 2-A testing.

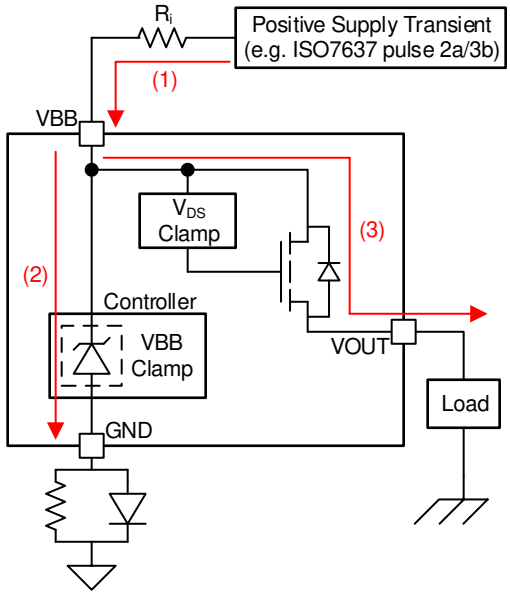


Figure 38. Current Path During Supply Voltage Transient

Feature Description (continued)

9.3.1.3.1 Load Dump

The TPS2HB35-Q1 device is tested according to ISO 16750-2:2010(E) suppressed load dump pulse. The device supports up to 40-V load dump transient and will maintain normal operation during the load dump pulse. If the switch is enabled, it will stay enabled and if the switch is disabled, it will stay disabled.

9.3.1.4 Driving Inductive Loads

When switching off an inductive load, the inductor may impose a negative voltage on the output of the switch. The TPS2HB35-Q1 includes a voltage clamp to limit voltage across the FET. The maximum acceptable load inductance is a function of the device robustness.

For more information on driving inductive loads, refer to TI's *How To Drive Inductive, Capacitive, and Lighting Loads with Smart High Side Switches* application report.

9.3.1.5 Reverse Battery

In the reverse battery condition, the switch will automatically be enabled regardless of the state of EN1/EN2 to prevent excess power dissipation inside the MOSFET body diode. In many applications (for example, resistive loads), the full load current may be present during reverse battery. In order to activate the automatic switch on feature, the SEL2 pin must have a path to module ground. This may be path 1 as shown in Figure 39, or if the SEL2 pin is unused, the path may be through R_{PROT} to module ground.

Protection features like thermal shutdown are not available during a reverse battery event. Care must be taken to ensure that excessive power is not dissipated in the switch during the reverse battery condition.

There are two options for blocking reverse current in the system. The first option is to place a blocking device (FET or diode) in series with the battery supply, blocking all current paths. The second option is to place a blocking diode in series with the GND node of the high-side switch. This method will protect the controller portion of the switch (path 2), but it will not prevent current from flowing through the load (path 3). The diode used for the second option may be shared amongst multiple high-side switches.

Path 1 shown in Figure 39 is blocked inside of the device.

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Feature Description (continued)

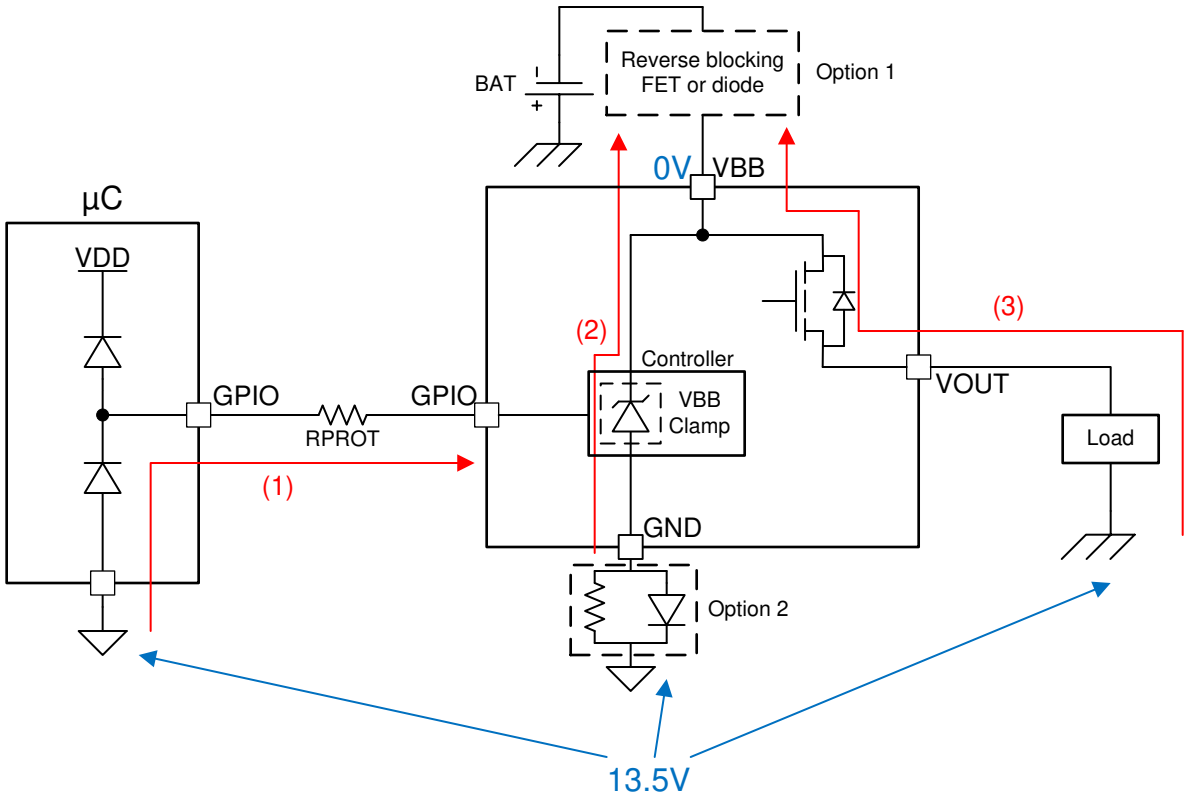


Figure 39. Current Path During Reverse Battery

9.3.1.6 Fault Event – Timing Diagrams (Version A/B/C)

NOTE

All timing diagrams assume that the SELx pins are set to select the relevant channel.

The LATCH, DIA_EN, and ENx pins are controlled by the user. The timing diagrams represent a possible use-case.

Figure 40 shows the immediate current limit switch off and the retry behavior of versions A and B of the device. As shown, the switch will remain latched off until the LATCH pin is low.

Feature Description (continued)

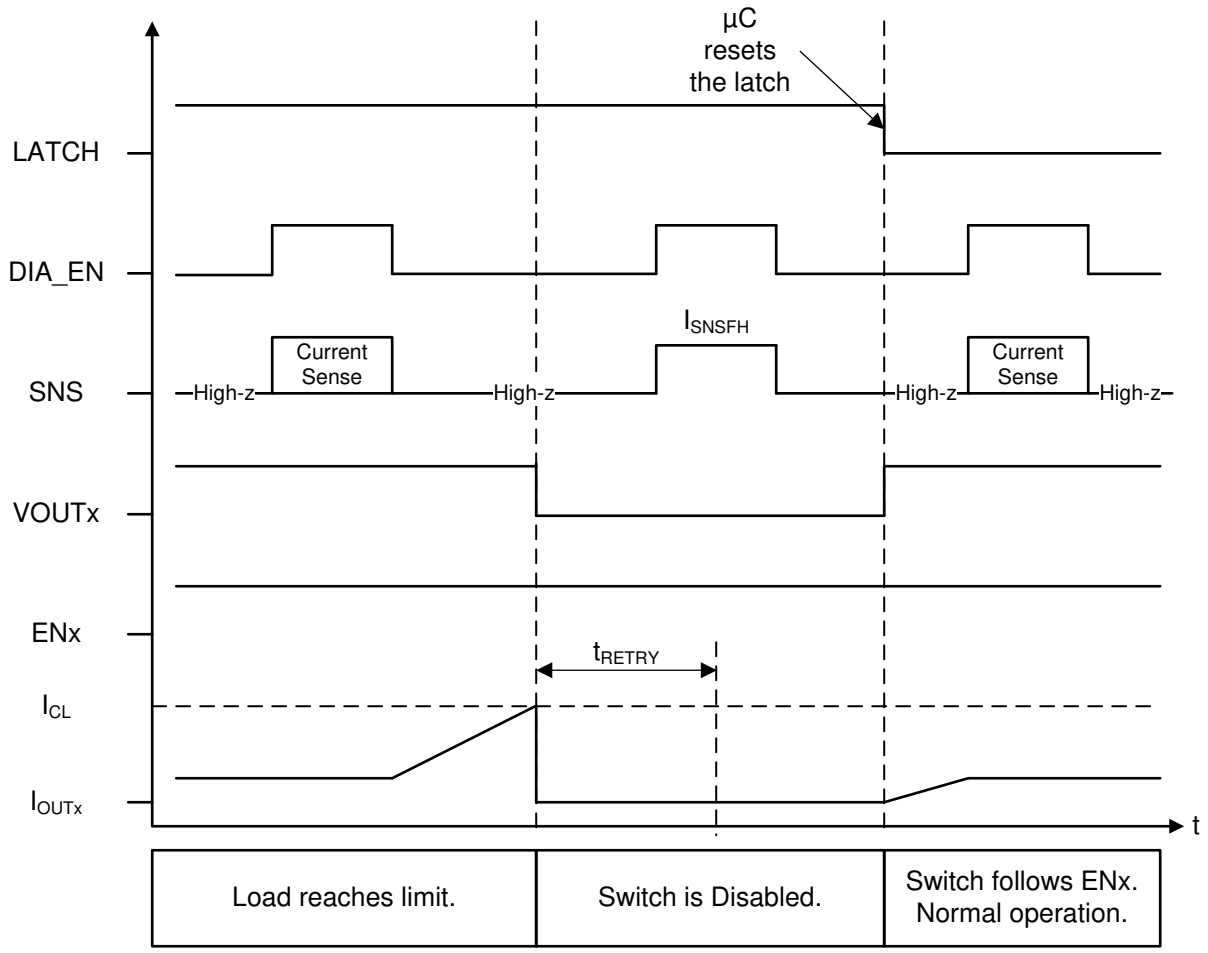


Figure 40. Current Limit – Version A and B - Latched Behavior

Figure 41 shows the immediate current limit switch off behavior of versions A and B. In this example, LATCH is tied to GND; hence, the switch will retry after the fault is cleared and t_{RETRY} has expired.

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Feature Description (continued)

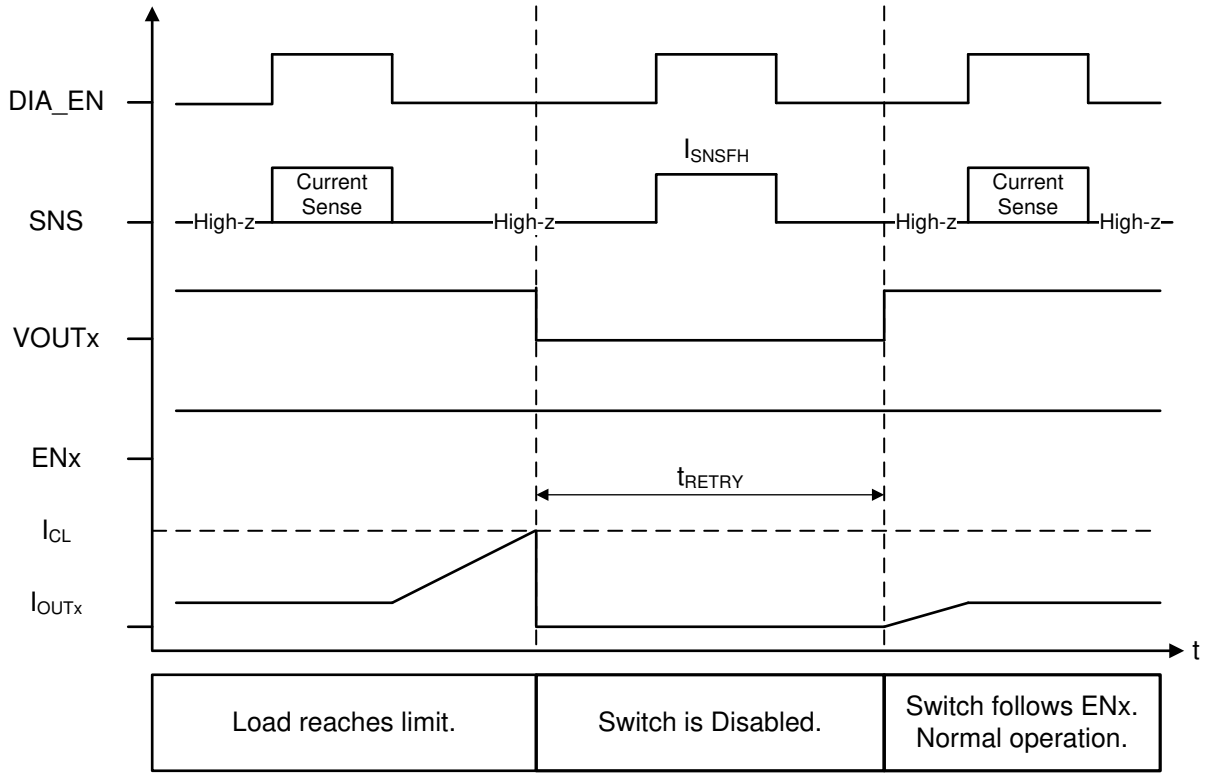


Figure 41. Current Limit – Version A and B - LATCH = 0

Figure 42 shows the active current limiting behavior of version C. In version C, the switch will not shutdown until thermal shutdown is reached.

Feature Description (continued)

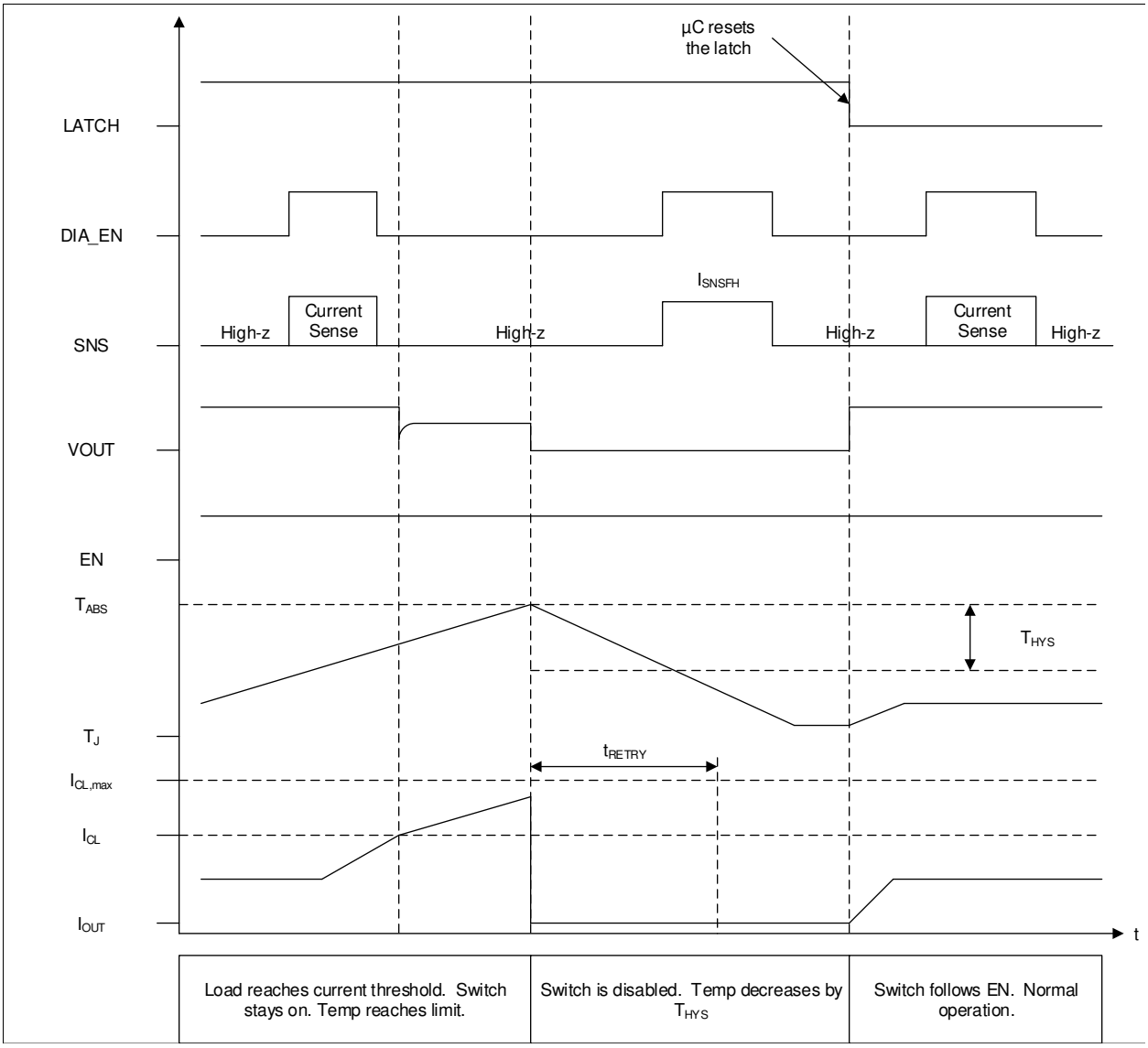


Figure 42. Current Limit – Version C - Latched Behavior

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Feature Description (continued)

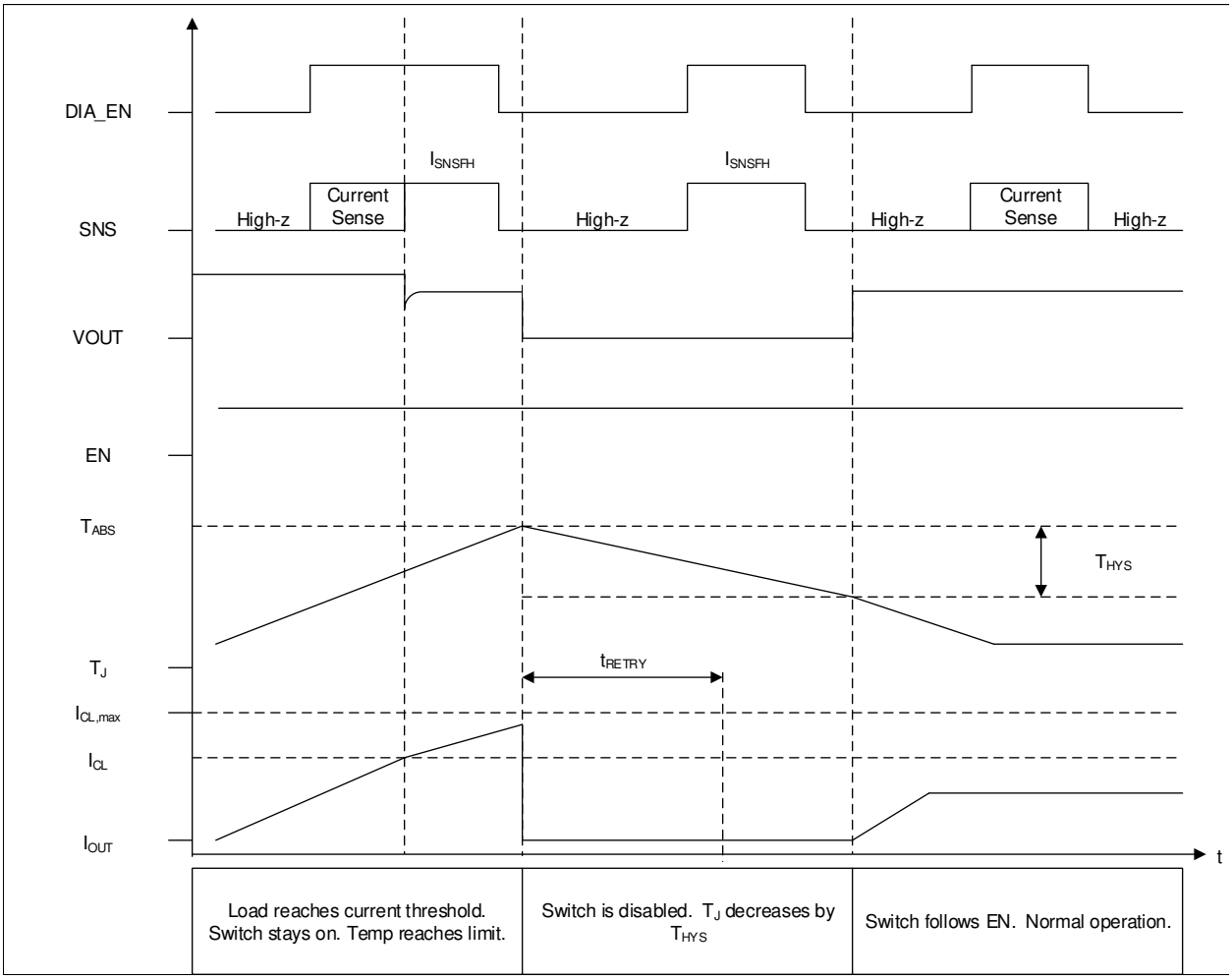


Figure 43. Current Limit – Version C - LATCH = 0

Figure 44 illustrates auto-retry behavior and provides a zoomed-in view of the fault indication during retry. When the switch retries after a shutdown event, the SNS fault indication will remain at the fault state until V_{OUT} has risen to V_{BB} – 1.8 V. Once V_{OUT} has risen, the SNS fault indication is reset and current sensing is available. If there is a short-to-ground and V_{OUT} cannot rise, the SNS fault indication will remain indefinitely.

NOTE

Figure 44 assumes that t_{RETRY} has expired by the time that T_J reaches the hysteresis threshold.

LATCH = 0 V and DIA_EN = 5 V

Feature Description (continued)

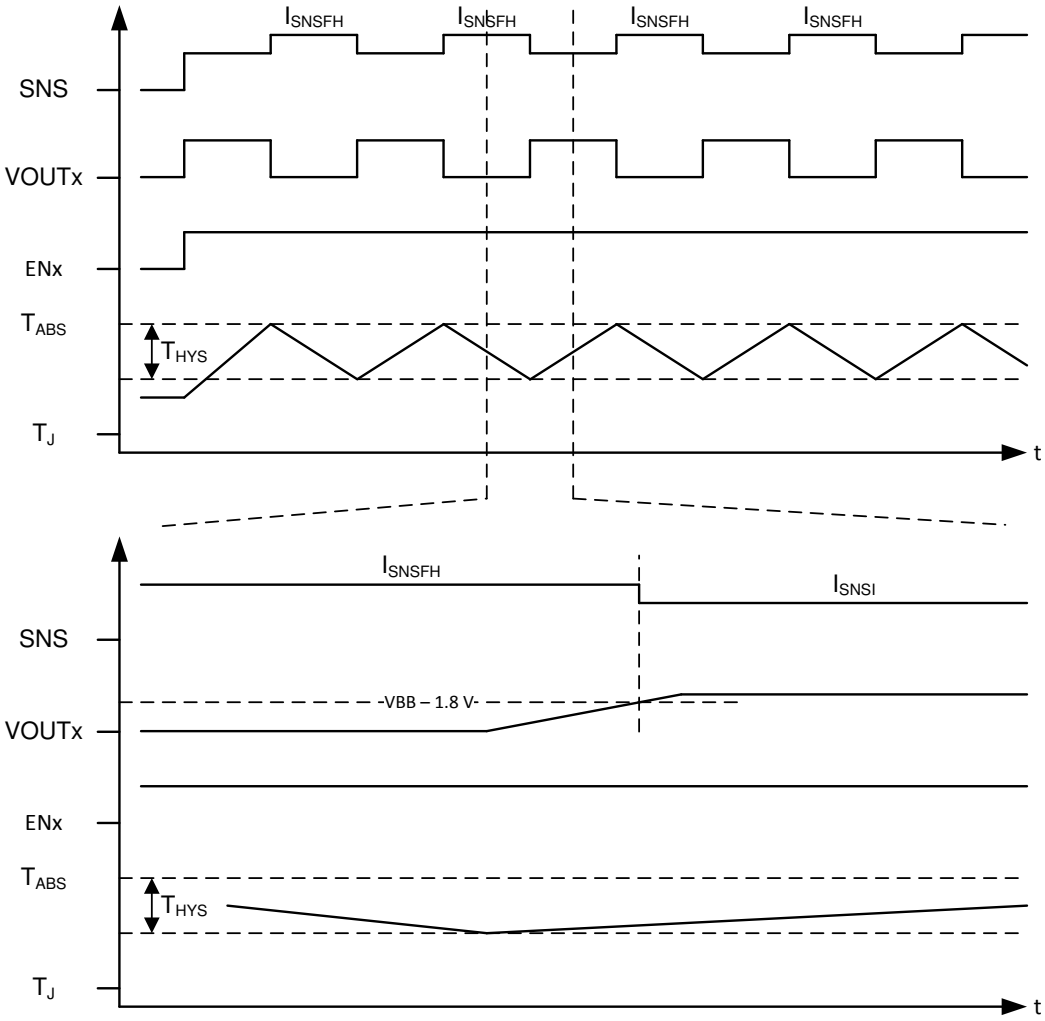


Figure 44. Fault Indication During Retry

9.3.2 Diagnostic Mechanisms

9.3.2.1 VOUTx Short-to-Battery and Open-Load

The TPS2HB35-Q1 is capable of detecting short-to-battery and open-load events regardless of whether the switch is turned on or off, however the two conditions use different methods.

9.3.2.1.1 Detection With Switch Enabled

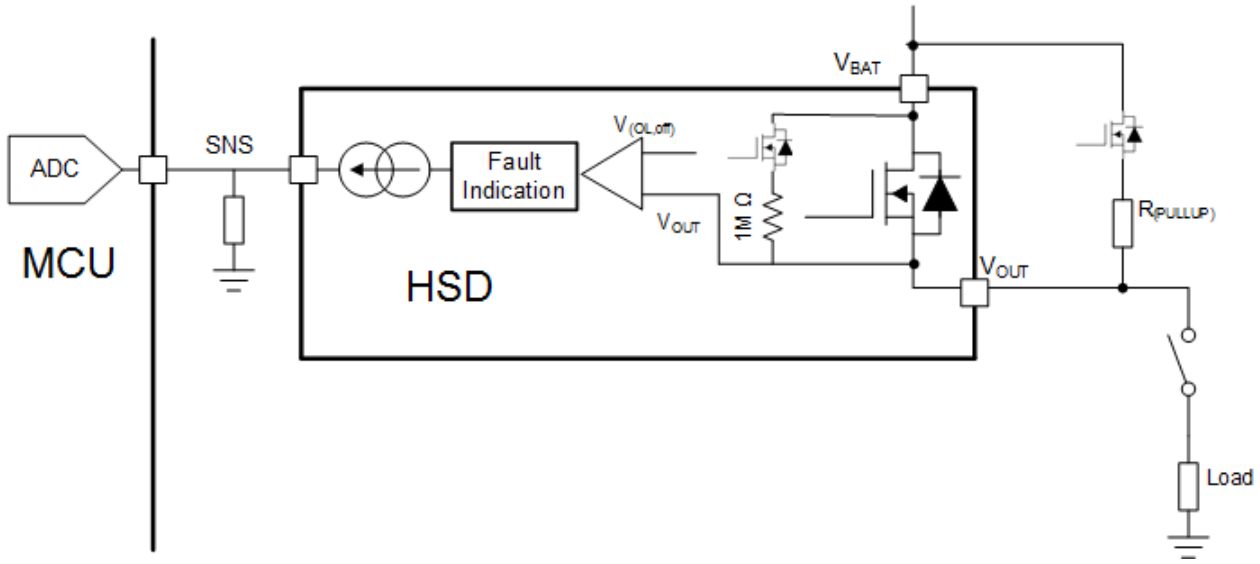
When the switch is enabled, the VOUTx short-to-battery and open-load conditions can be detected by the current sense feature. In both cases, the load current will be measured through the SNS pin as below the expected value.

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Feature Description (continued)

9.3.2.1.2 Detection With Switch Disabled

While the switch is disabled, if DIA_EN is high, an internal comparator will detect the condition of V_{OUT}. If the load is disconnected (open load condition) or there is a short to battery the V_{OUT} voltage will be higher than the open load threshold ($V_{OL,off}$) and a fault is indicated on the SNS pin. An internal pull-up of 1 MΩ is in series with an internal MOSFET switch, so no external component is required if only a completely open load must be detected. However, if there is significant leakage or other current draw even when the load is disconnected, a lower value pull-up resistor and switch can be added externally to set the V_{OUT} voltage above the $V_{OL,off}$ during open load conditions.



This figure assumes that the device ground and the load ground are at the same potential. In a real system, there may be a ground shift voltage of 1 V to 2 V.

Figure 45. Short to Battery and Open Load Detection

The detection circuitry is only enabled when DIA_EN = HIGH and EN = LOW. If $V_{OUT} > V_{OL}$, the SNS pin will go to the fault level, but if $V_{OUT} < V_{OL}$ there will be no fault indication. The fault indication will only occur if the SEL2 pin is set to diagnose the respective channel.

While the switch is disabled and DIA_EN is high, the fault indication mechanisms will continuously represent the present status. For example, if V_{OUT} decreases from greater than V_{OL} to less than V_{OL}, the fault indication is reset. Additionally, the fault indication is reset upon the falling edge of DIA_EN or the rising edge of EN.

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Feature Description (continued)

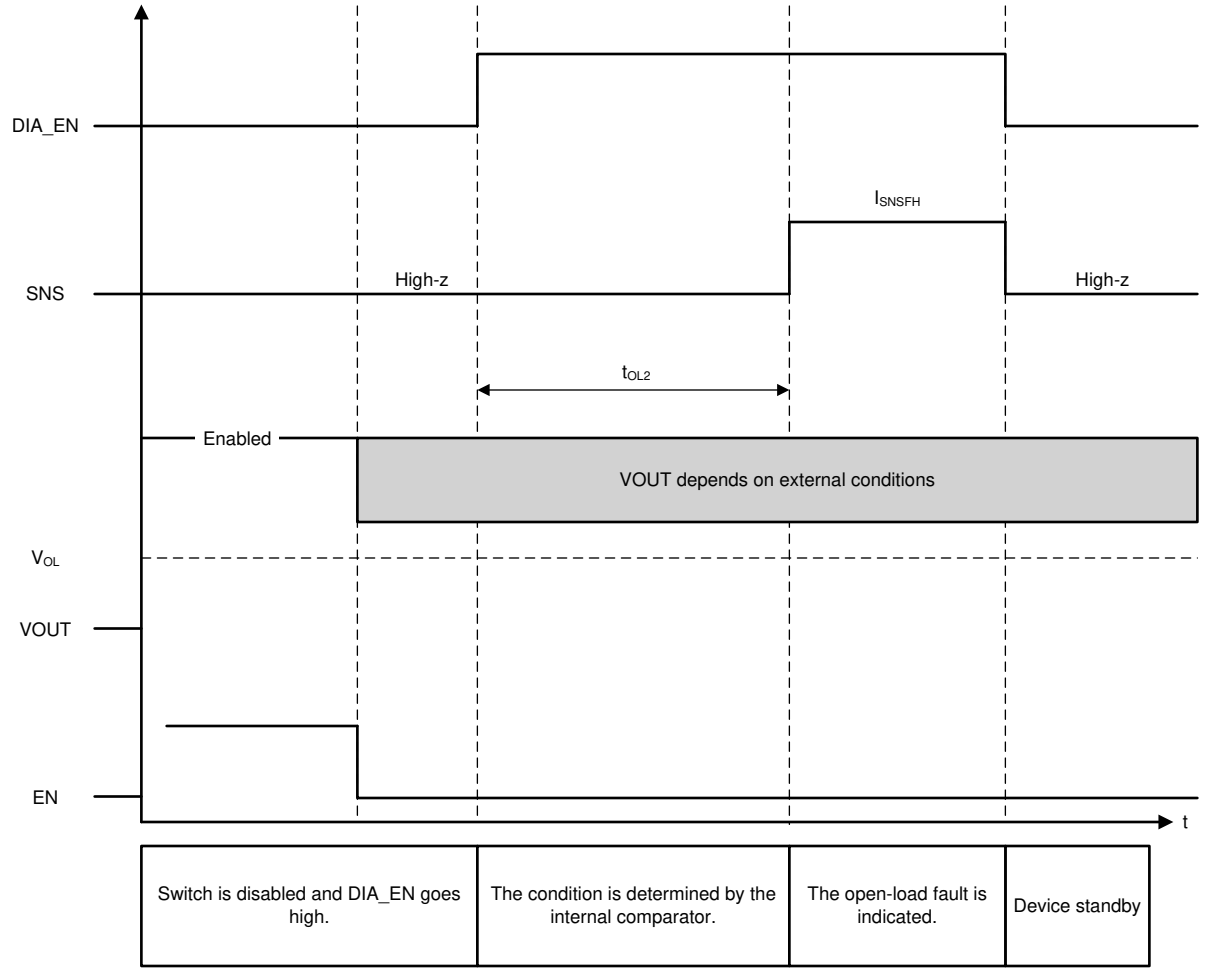


Figure 46. Open Load

9.3.2.2 SNS Output

The SNS output may be used to sense the load current or device temperature. The SELx pins will select the desired sense signal. The sense circuit will provide a current that is proportional to the selected parameter. This current will be sourced into an external resistor to create a voltage that is proportional to the selected parameter. This voltage may be measured by an ADC or comparator.

To ensure accurate sensing measurement, the sensing resistor should be connected to the same ground potential as the μC ADC.

Table 3. Analog Sense Transfer Function

PARAMETER	TRANSFER FUNCTION
Load current	$I_{SNSI} = I_{OUT} / K_{SNS}$ (2000)
Device temperature	$I_{SNST} = (T_J – 25°C) \times dI_{SNST} / dT + 0.85$

The SNS output will also be used to indicate system faults. I_{SNS} will go to the predefined level, I_{SNSFH}, when there is a fault. I_{SNSFH}, dI_{SNST}/dT, and K_{SNS} are defined in the *Specifications* section.

9.3.2.2.1 R_{SNS} Value

The following factors should be considered when selecting the R_{SNS} value:

- Current sense ratio (K_{SNS})

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- Largest and smallest diagnosable load current required for application operation
- Full-scale voltage of the ADC
- Resolution of the ADC

For an example of selecting R_{SNS} value, reference *Selecting the R_{SNS} Value* in the applications section of this datasheet.

9.3.2.2.1.1 High Accuracy Load Current Sense

In many automotive modules, it is required that the high-side switch provide diagnostic information about the downstream load. With more complex loads, high accuracy sensing is required. A few examples follow:

- **LED lighting**: In many architectures, the body control module (BCM) must be compatible with both incandescent bulbs and also LED modules. The bulb may be relatively simple to diagnose. However, the LED module will consume less current and also can include multiple LED strings in parallel. The same BCM is used in both cases, so the high-side switch can accurately diagnose both load types.
- **Solenoid protection**: Often solenoids are precisely controlled by low-side switches. However, in a fault event, the low-side switch cannot disconnect the solenoid from the power supply. A high-side switch can be used to continuously monitor several solenoids. If the system current becomes higher than expected, the high-side switch can disable the module.

9.3.2.2.1.2 SNS Output Filter

To achieve the most accurate current sense value, it is recommended to filter the SNS output. There are two methods of filtering:

- Low-Pass RC filter between the SNS pin and the ADC input. This filter is illustrated in Figure 50 with typical values for the resistor and capacitor. The designer should select a C_{SNS} capacitor value based on system requirements. A larger value will provide improved filtering but a smaller value will allow for faster transient response.
- The ADC and microcontroller can also be used for filtering. It is recommended that the ADC collects several measurements of the SNS output. The median value of this data set should be considered as the most accurate result. By performing this median calculation, the microcontroller can filter out any noise or outlier data.

9.3.2.3 Fault Indication and SNS Mux

The following faults will be communicated through the SNS output:

- Switch shutdown, due to:
  - Thermal Shutdown
  - Current limit
- Active current limiting
- Open-Load / V_{OUT} shorted-to-battery

Open-load and short-to-battery are not indicated while the switch is enabled, although these conditions can still be detected through the sense current. Hence, if there is a fault indication corresponding to an enabled channel, then it must be either due to an overcurrent or overtemperature event.

The SNS pin will only indicate the fault if the SELx pins are selecting the relevant channel. When the device is set to measure temperature, the pin will be measuring the temperature of whichever channel is at a higher temperature.

Table 4. Version A/B/C SNS Mux

INPUTS				OUTPUTS
DIA_EN	SEL1	SEL2	FAULT DETECT(1)	SNS
0	X	X	X	High-Z
1	0	0	0	CH1 current

(1) Fault Detect encompasses multiple conditions:
(a) Switch shutdown and waiting for retry
(b) Active current limiting (Version C)
(c) Open-load and short-to-battery

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Table 4. Version A/B/C SNS Mux (continued)

INPUTS				OUTPUTS
DIA_EN	SEL1	SEL2	FAULT DETECT(1)	SNS
1	0	1	0	CH2 current
1	1	0	0	Device temperature
1	1	1	0	N/A
1	0	0	1	I_{SNSFH}
1	0	1	1	I_{SNSFH}
1	1	0	1	Device temperature
1	1	1	1	N/A

9.3.2.4 Resistor Sharing

Multiple high-side channels may use the same SNS resistor as shown in Figure 47. This reduces the total number of passive components in the system and the number of ADC terminals that are required of the microcontroller.

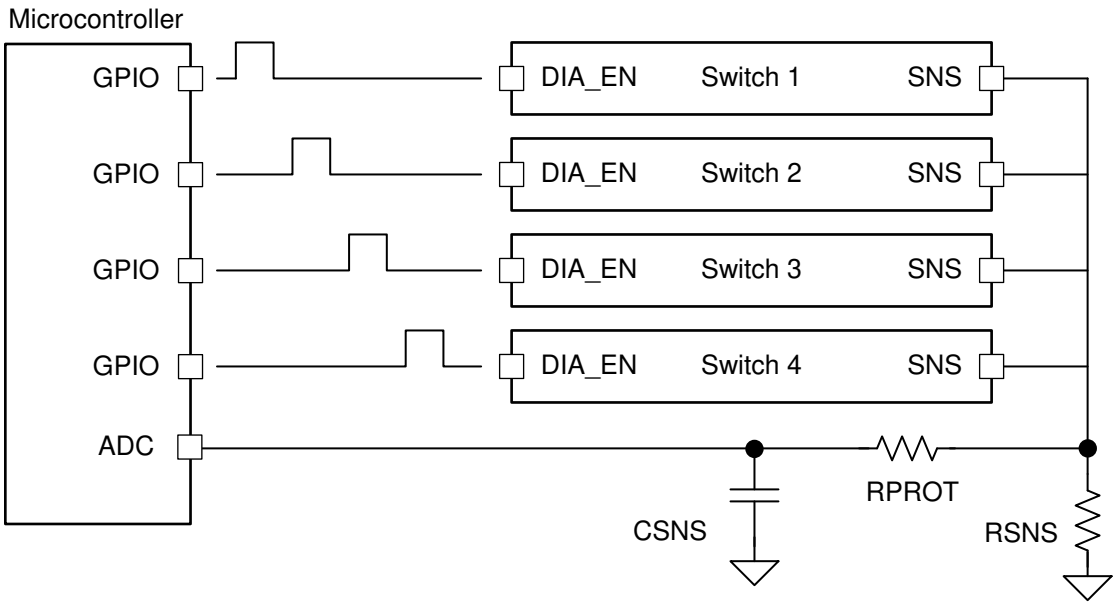


Figure 47. Sharing R_{SNS} Among Multiple Devices

9.3.2.5 High-Frequency, Low Duty-Cycle Current Sensing

Some applications will operate with a high-frequency, low duty-cycle PWM or require fast settling of the SNS output. For example, a 250 Hz, 5% duty cycle PWM will have an on-time of only 200 µs that must be accommodated. The micro-controller ADC may sample the SNS signal after the defined settling time $t_{SNSION3}$.

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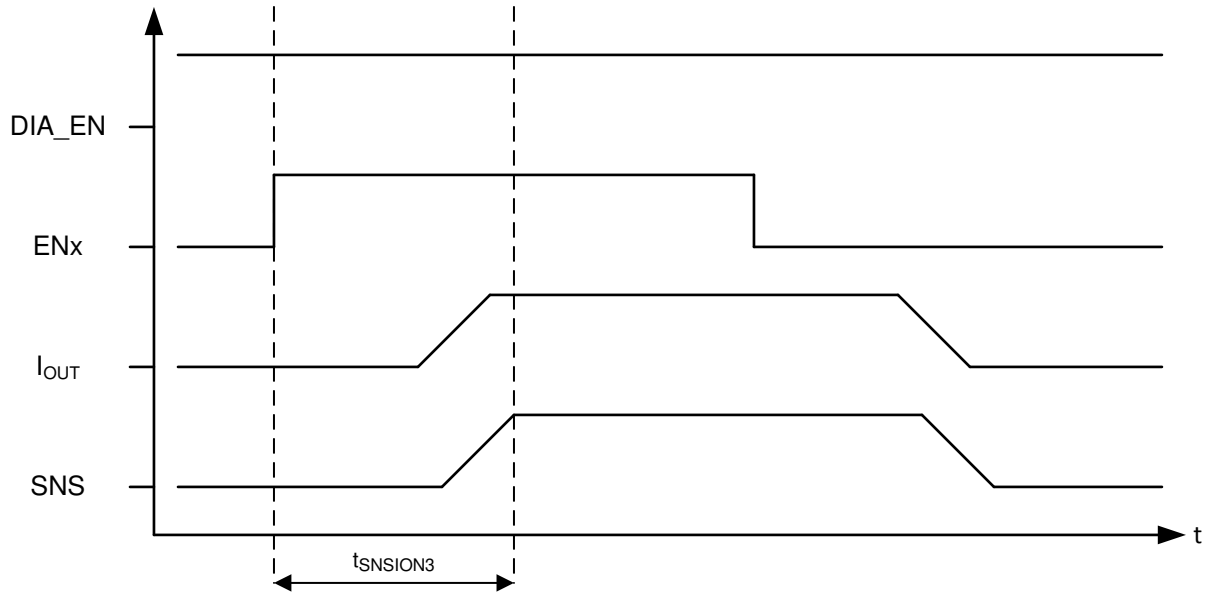


Figure 48. Current Sensing in Low-Duty Cycle Applications

9.4 Device Functional Modes

During typical operation, the TPS2HB35-Q1 can operate in a number of states that are described below and shown as a state diagram in Figure 49.

9.4.1 Off

Off state occurs when the device is not powered.

9.4.2 Standby

Standby state is a low-power mode used to reduce power consumption to the lowest level. Diagnostic capabilities are not available in Standby mode.

9.4.3 Diagnostic

Diagnostic state may be used to perform diagnostics while the switches are disabled.

9.4.4 Standby Delay

The Standby Delay state is entered when EN1, EN2, and DIA_EN are low. After t_{STBY}, if the ENx and DIA_EN pins are still low, the device will go to Standby state.

9.4.5 Active

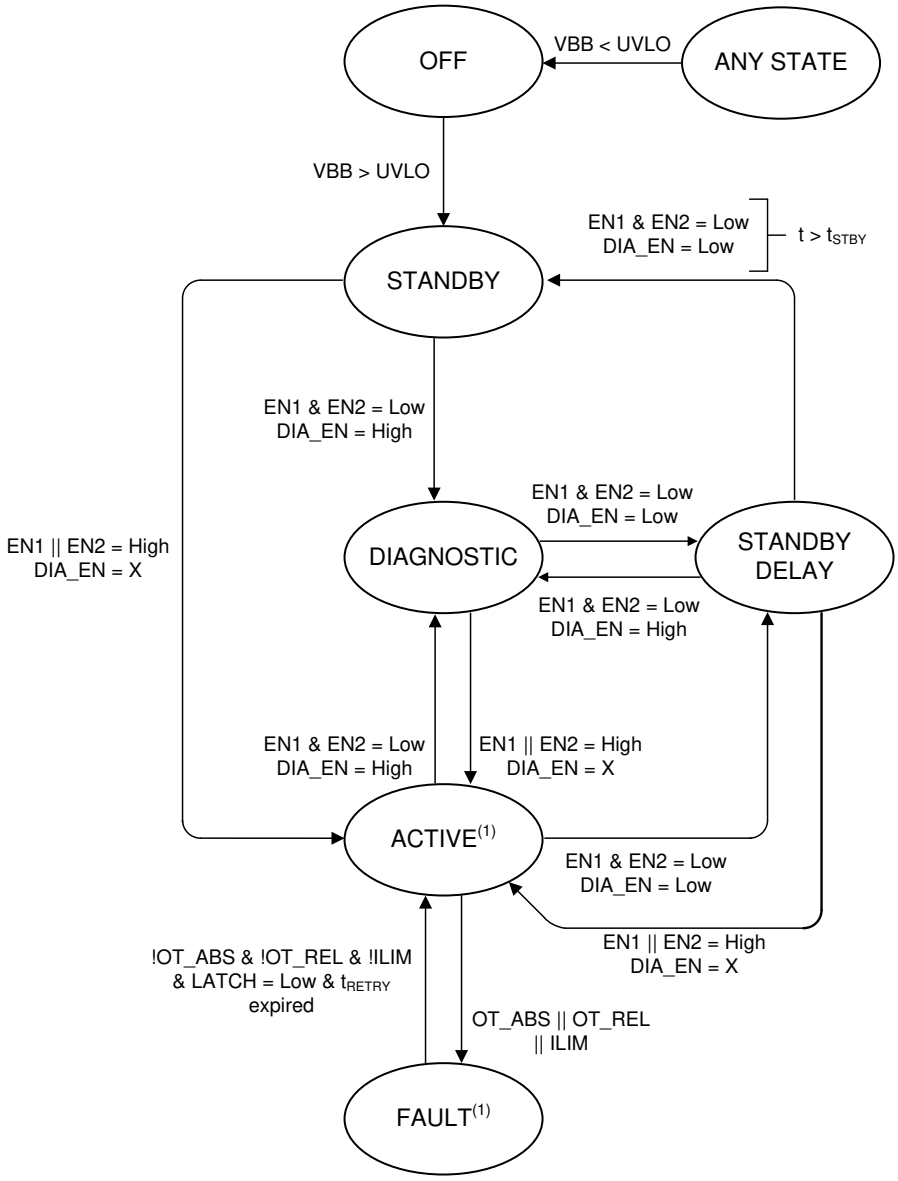
In Active state, one or more of the switches are enabled. The diagnostic functions may be turned on or off during Active state.

9.4.6 Fault

The Fault state is entered if a fault shutdown occurs (thermal shutdown or current limit). After all faults are cleared, the LATCH pin is low, and the retry timer has expired, the device will transition out of Fault state. If the relevant ENx pin is high, the switch will re-enable. If the relevant ENx pin is low, the switch will remain off.

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Device Functional Modes (continued)



CH1 and CH2 operate independently. Each channel is enabled or disabled independently. Also, if there is a fault on one channel, the other channel is not affected.

Figure 49. State Diagram

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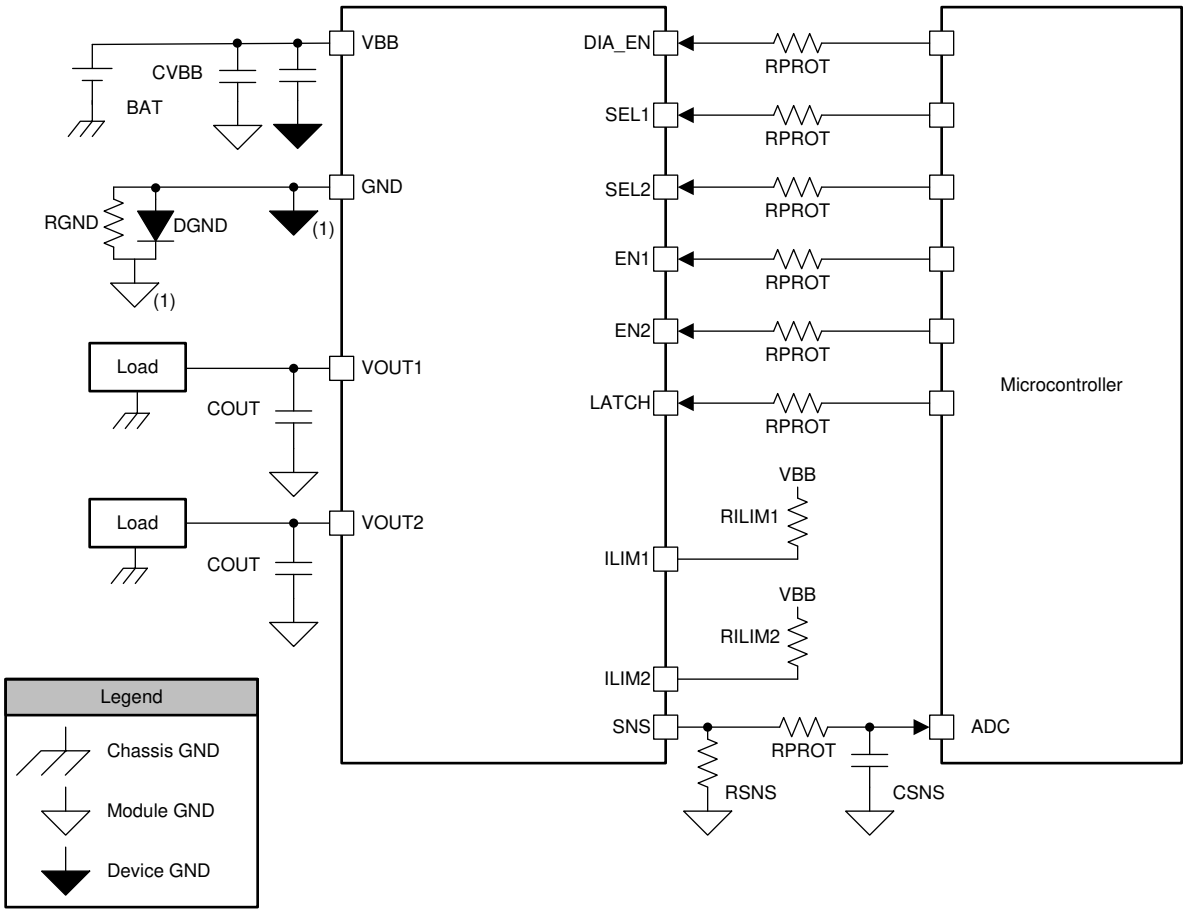
10 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

10.1 Application Information

Figure 50 shows the schematic of a typical application for the TPS2HB35-Q1. It includes all standard external components. This section of the datasheet discusses the considerations in implementing commonly required application functionality.



With the ground protection network, the device ground will be offset relative to the microcontroller ground.

Figure 50. System Diagram

Table 5. Recommended External Components

COMPONENT	TYPICAL VALUE	PURPOSE
R_{PROT}	15 kΩ	Protect the microcontroller and device I/O pins.
R_{SNS}	1 kΩ	Translate the sense current into sense voltage.
C_{SNS}	100 pF - 10 nF	Creates Low-pass filter for the ADC input
R_{GND}	4.7 kΩ	Stabilize GND potential during turn-off of inductive load.
D_{GND}	BAS21 Diode	Protect the device during reverse battery.

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Application Information (continued)

Table 5. Recommended External Components (continued)

COMPONENT	TYPICAL VALUE	PURPOSE
R_{ILIM}	5 kΩ - 25 kΩ	Set the current limit threshold.
C_{VBB}	4.7 nF to Device GND	Filters voltage transients (for example, ESD, ISO7637-2) and improves emissions
	220 nF to Module GND	Stabilize the input supply and filter out low frequency noise.
C_{OUT}	220 nF	Filters voltage transients (for example, ESD, ISO7637-2)

10.1.1 Ground Protection Network

As discussed in the *Reverse Battery* section, D_{GND} may be used to prevent excessive reverse current from flowing into the device during a reverse battery event. Additionally, R_{GND} is placed in parallel with D_{GND} if the switch is used to drive an inductive load. The ground protection network (D_{GND} and R_{GND}) may be shared amongst multiple high-side switches.

A minimum value for R_{GND} may be calculated by using the absolute maximum rating for I_{GND}. During the reverse battery condition, $I_{GND} = V_{BB} / R_{GND}$:

$R_{GND} \geq V_{BB} / I_{GND}$

- Set V_{BB} = –13.5 V
- Set I_{GND} = –50 mA (absolute maximum rating)

$$R_{GND} \geq –13.5\ V / –50\ mA = 270\ \Omega \quad (2)$$

In this example, it is found that R_{GND} must be at least 270 Ω. It is also necessary to consider the power dissipation in R_{GND} during the reverse battery event:

$$P_{RGND} = V_{BB}^2 / R_{GND} \quad (3)$$

$P_{RGND} = (13.5\ V)^2 / 270\ \Omega = 0.675\ W$

In practice, R_{GND} may not be rated for such a high power. In this case, a larger resistor value should be selected.

10.1.2 Interface With Microcontroller

The ground protection network will cause the device ground to be at a higher potential than the module ground (and microcontroller ground). This offset will impact the interface between the device and the microcontroller.

Logic pin voltage will be offset by the forward voltage of the diode. For input pins (for example, EN1), the designer must consider the V_{IH} specification of the switch and the V_{OH} specification of the microcontroller. For a system that *does not* include D_{GND}, it is required that $V_{OH} > V_{IH}$. For a system that *does* include D_{GND}, it is required that $V_{OH} > (V_{IH} + V_F)$. V_F is the forward voltage of D_{GND}.

The sense resistor, R_{SNS}, should be terminated to the microcontroller ground. In this case, the ADC can accurately measure the SNS signal even if there is an offset between the microcontroller ground and the device ground.

10.1.3 I/O Protection

R_{PROT} is used to protect the microcontroller I/O pins during system-level voltage transients such as ISO pulses or reverse battery. The SNS pin voltage can exceed the ADC input pin maximum voltage if the fault or saturation current causes a high enough voltage drop across the sense resistor. If that can occur in the design (for example, by switching to a high value R_{SNS} to improve ADC input level), then an appropriate external clamp has to be designed to prevent a high voltage at the SNS output and the ADC input.

10.1.4 Inverse Current

Inverse current occurs when 0 V < V_{BB} < V_{OUTx}. In this case, current may flow from V_{OUTx} to V_{BB}. Inverse current cannot be caused by a purely resistive load. However, a capacitive or inductive load can cause inverse current. For example, if there is a significant amount of load capacitance and the V_{BB} node has a transient droop, V_{OUTx} may be greater than V_{BB}.

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The TPS2HB35-Q1 will not detect inverse current. When the switch is enabled, inverse current will pass through the switch. When the switch is disabled, inverse current may pass through the MOSFET body diode. The device will continue operating in the normal manner during an inverse current event.

10.1.5 Loss of GND

The ground connection may be lost either on the device level or on the module level. If the ground connection is lost, both switches will be disabled. If the switch was already disabled when the ground connection was lost, the switch will remain disabled. When the ground is reconnected, normal operation will resume.

10.1.6 Automotive Standards

The TPS2HB35-Q1 is designed to be protected against all relevant automotive standards to ensure reliable operations when connected to a 12-V automotive battery.

10.1.6.1 ISO7637-2

The TPS2HB35-Q1 is tested according to the ISO7637-2:2011 (E) standard. The test pulses are applied both with the switches enabled and disabled. The test setup includes only the DUT and minimal external components: C_{VBB}, C_{OUT}, D_{GND}, and R_{GND}.

Status II is defined in ISO 7637-1 Function Performance Status Classification (FPSC) as: *The function does not perform as designed during the test but returns automatically to normal operation after the test*. See Table 6 for ISO7637-2:2011 (E) expected results.

Table 6. ISO7637-2:2011 (E) Results

TEST PULSE	TEST PULSE SEVERITY LEVEL WITH STATUS II FUNCTIONAL PERFORMANCE		MINIMUM NUMBER OF PULSES OR TEST TIME	BURST CYCLE / PULSE REPETITION TIME	
	LEVEL	US		MIN	MAX
1	III	–112 V	500 pulses	0.5 s	--
2a(1)	III	+55 V	500 pulses	0.20 s	5 s
2b	IV	+10 V	10 pulses	0.5 s	5 s
3a	IV	–220 V	1 hour	90 ms	100 ms
3b	IV	+150 V	1 hour	90 ms	100 ms

(1) 1 µF capacitance on C_{VBB} is required for passing level 3 ISO7637 pulse 2 A.

10.1.6.2 AEC – Q100-012 Short Circuit Reliability

The TPS2HB35-Q1 is tested according to the AEC-Q100-012 Short Circuit Reliability standard. This test is performed to demonstrate the robustness of the device against V_{OUT} short-to-ground events. Test conditions and test procedures are summarized in Table 7. For further details, refer to the AEC - Q100-012 standard document.

Test conditions:

- LATCH = 0 V
- I_{LIM} = 5 kΩ
- 10 units from 3 separate lots for a total of 30 units
- L_{supply} = 5 µH, R_{supply} = 10 mΩ
- V_{BB} = 14 V

Test procedure:

- Parametric data is collected on each unit pre-stress.
- Each unit is enabled into a short-circuit with the required short circuit cycles or duration as specified.
- Functional testing is performed on each unit post-stress to verify that the part still operates as expected.

The cold repetitive test is run at 85ºC which is the worst case condition for the device to sustain a short circuit. The cold repetitive test refers to the device being given time to cool down between pulses, rather than being run at a cold temperature. The load short circuit is the worst case situation, since the energy stored in the cable inductance can cause additional harm. The fast response of the device ensures current limiting occurs quickly and at a current close to the load short condition. In addition, the hot repetitive test is performed as well.

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Table 7. AEC - Q100-012 Test Results

TEST	LOCATION OF SHORT	DEVICE VERSION	NO. OF CYCLES / DURATION	NO. OF UNITS	NO. OF FAILS
Cold Repetitive - Long Pulse	Load Short Circuit, L_{short} = 5 µH, R_{short} = 100 mΩ, T_A = 85ºC	B	100 k cycles	30	0
Hot Repetitive - Long Pulse	Load Short Circuit, L_{short} = 5 µH, R_{short} = 100 mΩ, T_A = 25ºC	B	100 hours	30	0

10.2 Typical Application

This application example demonstrates how the TPS2HB35-Q1 device can be used to power resistive heater loads in automotive seats. In this example, we consider dual heater loads that are powered independently by the two channels of the device. A dual-channel device is the ideal solution as it will yield a smaller solution size relative to two single-channel devices.

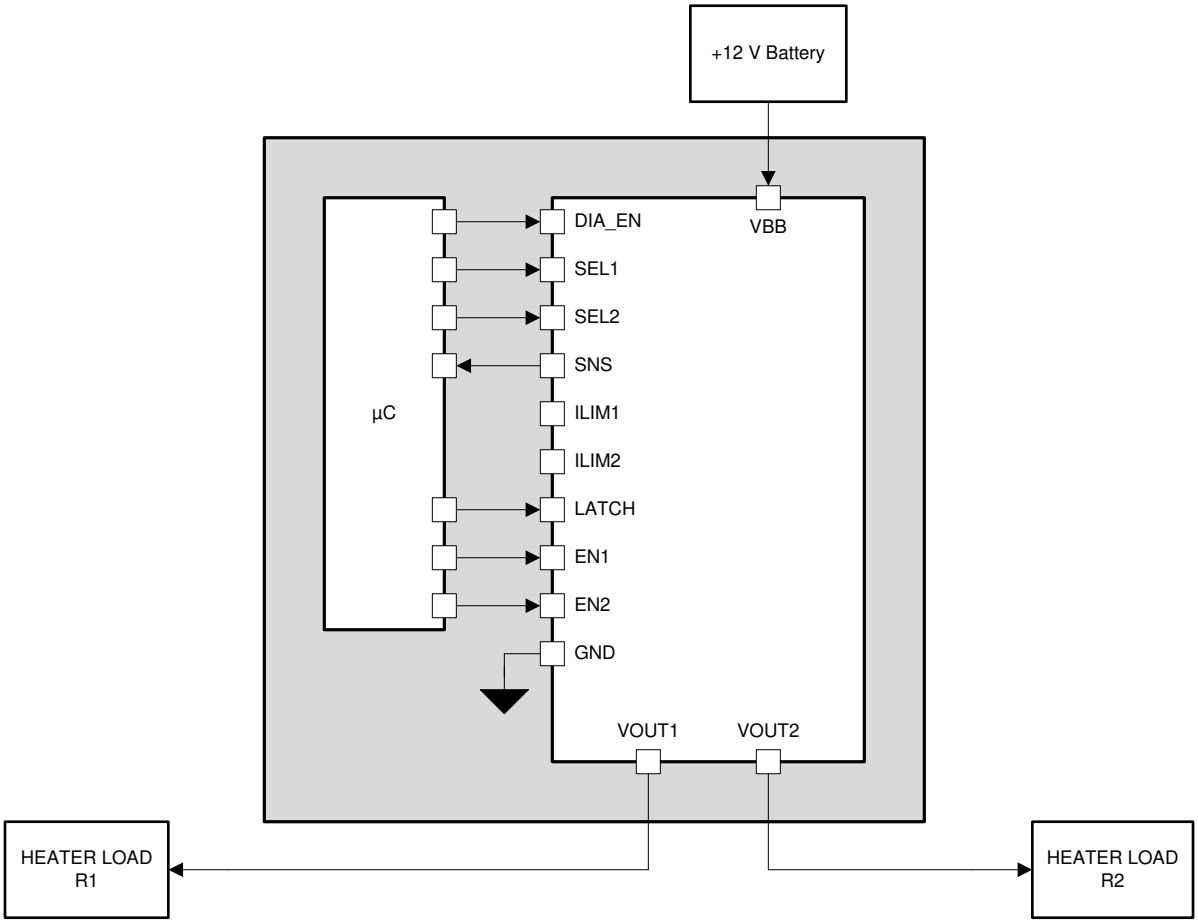


Figure 51. Block Diagram for Powering Dual Heater Loads

10.2.1 Design Requirements

For this design example, use the input parameters shown in Table 8.

Table 8. Design Parameters

DESIGN PARAMETER	EXAMPLE VALUE
V_{BB}	13.5 V
Load Ch1 - Heater 1	40 W max
Load Ch2 - Heater 2	40 W max

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Table 8. Design Parameters (continued)

DESIGN PARAMETER	EXAMPLE VALUE
Load Current Sense	40 mA to 8 A
I_{LIM}	5 A
Ambient temperature	70°C
$R_{\theta JA}$	32.9°C/W (depending on PCB)
Device Version	A

10.2.2 Detailed Design Procedure

10.2.2.1 Thermal Considerations

The DC current in each channel under maximum load power condition will be around 3 A. Both heater loads can be ON at the same time, so the case where both channels are enabled simultaneously is considered to assume worst case heating.

Power dissipation in the switch is calculated in Equation 4. R_{ON} is assumed to be 75 mΩ because this is the maximum specification at high temperature. In practice, R_{ON} will almost always be lower.

$$P_{FET} = I^2 \times R_{ON} \quad (4)$$

$$P_{FET} = (3\ A)^2 \times 75\ m\Omega = 0.675\ W \quad (5)$$

If both channels are enabled, then the total power dissipation is 1.35 W. The junction temperature of the device can be calculated using Equation 6 and the $R_{\theta JA}$ value from the *Specifications* section.

$$T_J = T_A + R_{\theta JA} \times P_{FET} \quad (6)$$

$$T_J = 70°C + 32.9°C/W \times 1.35\ W = 114.4°C$$

The maximum junction temperature rating for the TPS2HB35-Q1 device is T_J = 150°C. Based on the above example calculation, the device temperature will stay below the maximum rating.

10.2.2.2 R_{ILIM} Calculation

In this application, the TPS2HB35-Q1 must allow for the maximum 3-A current with margin but minimize the energy in the switch during a fault condition by minimizing the current limit. For this application, the best I_{LIM} set point is approximately 5 A. Equation 7 allows you to calculate the R_{ILIM} value that is placed from the I_{LIMX} pins to V_{BB}. R_{ILIM} is calculated in kΩ.

$$R_{ILIM} = K_{CL} / I_{CL} \quad (7)$$

Because this device is version A, the K_{CL} value in the *Specifications* section is 50 A × kΩ.

$$R_{ILIM} = 50\ A \times k\Omega / 5\ A = 10\ k\Omega \quad (8)$$

For a I_{LIM} of 5 A, the R_{ILIM} value should be set at approximately 10 kΩ.

10.2.2.3 Diagnostics

If the resistive heating load is disconnected (heater malfunction), an alert is desired. Open-load detection can be performed in the switch-enabled state with the current sense feature of the TPS2HB35-Q1 device. Under open load condition, the current in the SNS pin will be the fault current and this can be detected from the sense voltage measurement.

10.2.2.3.1 Selecting the R_{SNS} Value

Table 9 shows the requirements for the load current sense in this application. The K_{SNS} value is specified for the device and can be found in the *Specifications* section.

Table 9. R_{SNS} Calculation Parameters

PARAMETER	EXAMPLE VALUE
Current Sense Ratio (K_{SNS})	2000
Largest diagnosable load current	8 A
Smallest diagnosable load current	40 mA
Full-scale ADC voltage	5-V

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Table 9. R_{SNS} Calculation Parameters (continued)

PARAMETER	EXAMPLE VALUE
ADC resolution	10-bit

The load current measurement requirements of 8 A ensures that even in the event of a overcurrent surpassing the device internal 5-A limit, the MCU can register and react by shutting down the TPS2HB35-Q1, while the low level of 40 mA allows for accurate measurement of low load currents.

The R_{SNS} resistor value should be selected such that the largest diagnosable load current puts V_{SNS} at about 95% of the ADC full-scale. With this design, any ADC value above 95% can be considered a fault. Additionally, the R_{SNS} resistor value should ensure that the smallest diagnosable load current does not cause V_{SNS} to fall below 1 LSB of the ADC. With the given example values, a 1.2-kΩ sense resistor satisfies both requirements shown in Table 10.

Table 10. V_{SNS} Calculation

LOAD (A)	SENSE RATIO	I_{SNS} (mA)	R_{SNS} (Ω)	V_{SNS} (V)	% of 5-V ADC
0.040	2000	0.02	1200	0.024	0.5%
8	2000	4	1200	4.800	96.0%

10.2.3 Application Curves

When the device receives a rising edge on the ENx pulse the output will turn on as shown in Figure 52. After the turn-on delay time, the device V_{OUT} goes to the V_{BB} supply and begins outputting the steady state resistive current.

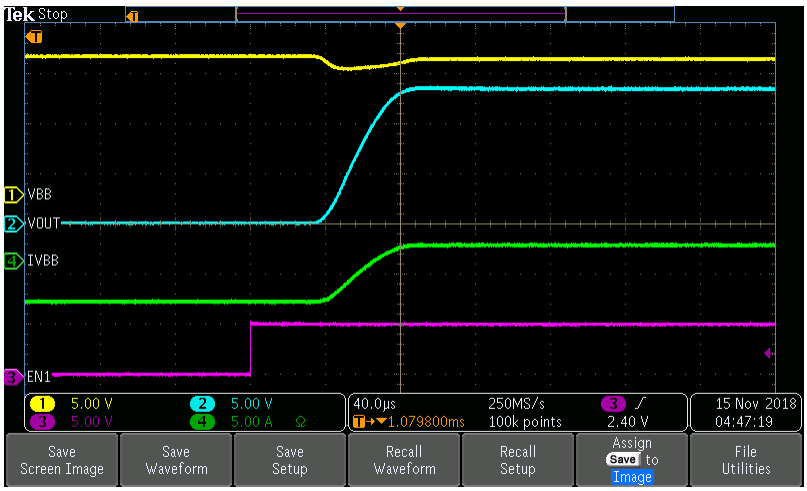


Figure 52. Turn-On Waveform

When the device turns off on a falling edge of ENx, the channel I_{OUT} will go to zero and the V_{OUT} will drop to zero as well as shown in Figure 53.

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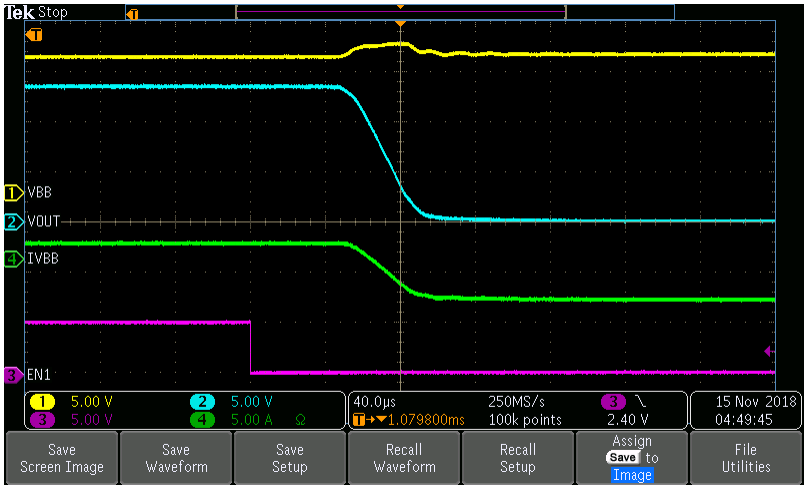


Figure 53. Turn-Off Waveform

While enabled, it is important to measure the output current through both channels. Figure 54 shows this behavior when toggling the SELx pins. The image shows that when SEL2 toggles high to low, the SNS pin toggles between representing I_{OUT1} and I_{OUT2}. When SEL2 is low SNS represents I_{OUT1} and when SEL2 is high SNS represents I_{OUT2}. This image shows that channel 2 is currently outputting twice the output current as channel 1.

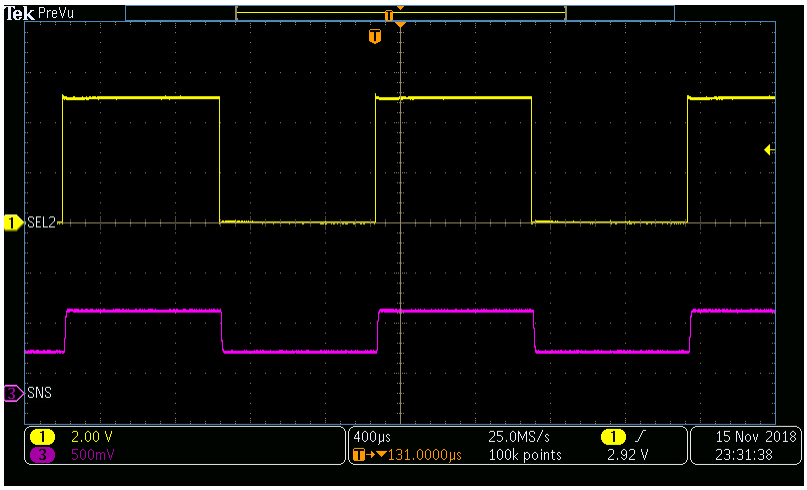


Figure 54. Toggling Between CH1 and CH2 Current Measurement

If the output of the TPS2HB35-Q1 is short-circuited, the device will protect the system from failure. shows the device turning off the output at a set current limit when the output is short circuited. (Note: shows a case with a higher R_{ILIM} than calculated in this example, so the current limit is higher than 8 A).

The TPS2HB35-Q1 also has a variant that allows the current to remain on when the current threshold is reached. This allow capacitors to be charged up in one attempt instead of hitting the current limit and immediately shutting down. The short circuit behavior can be seen in Figure 55

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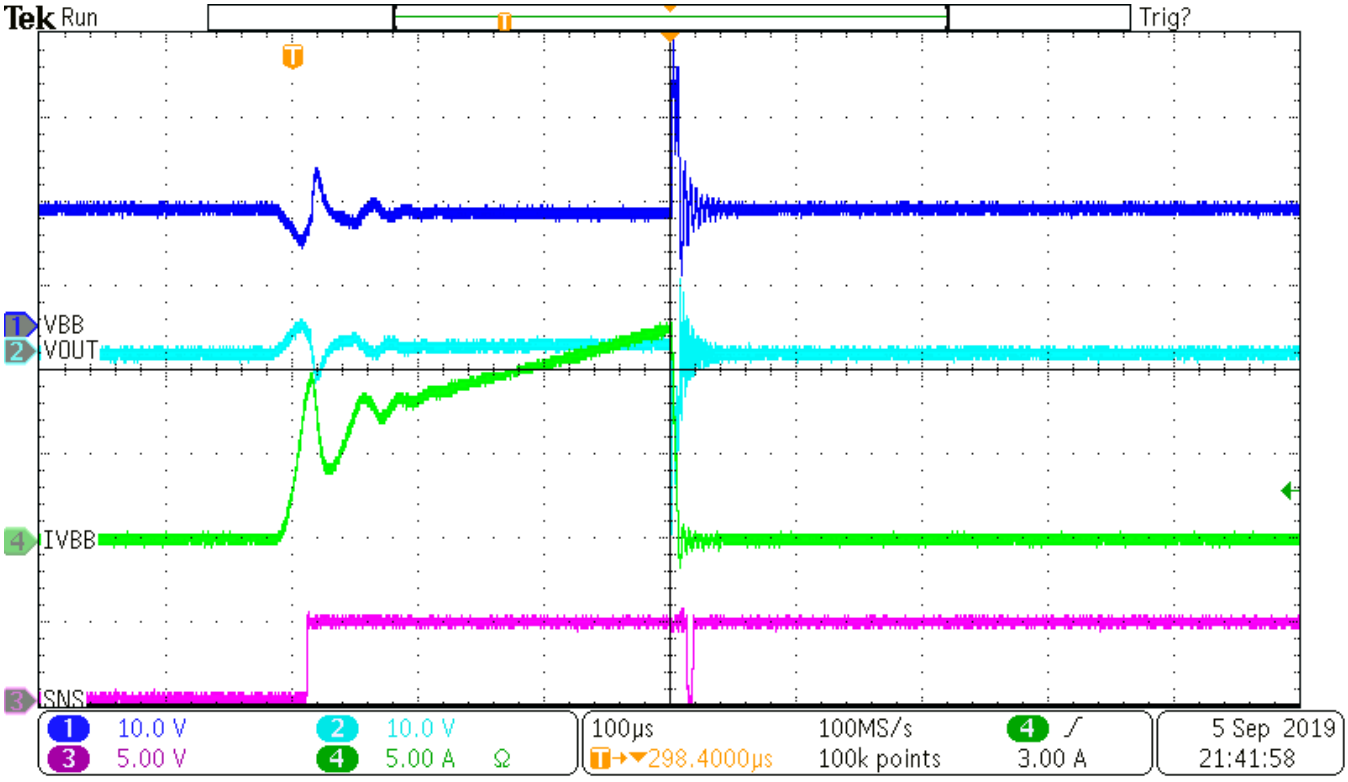


Figure 55. TPS2HB35C-Q1 Short Circuit Waveform

Figure 56 shows the TPS2HB35C-Q1 device charging up a 270-µF capacitor in parallel with a 6-Ω resistor to 16 V at 85ºC on both channels. The current waveform is the combined current going through both channels.

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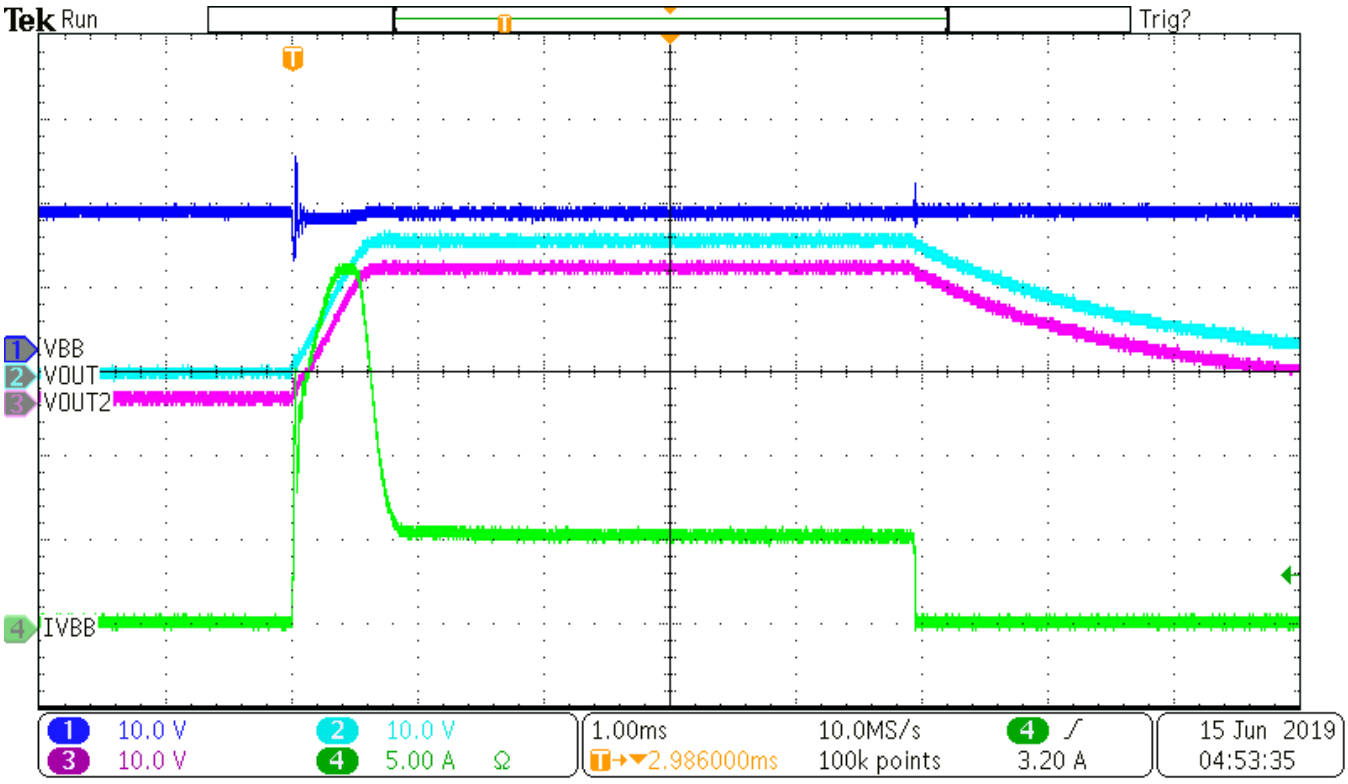


Figure 56. TPS2HB35C-Q1 Charging a 270-µF Capacitor

11 Power Supply Recommendations

The TPS2HB35-Q1 device is designed to operate in a 12-V automotive system. The nominal supply voltage range is 6 V to 18 V as measured at the V_{BB} pin with respect to the GND pin of the device. In this range the device meets full parametric specifications as listed in the *Electrical Characteristics* table. The device is also designed to withstand voltage transients beyond this range. When operating outside of the nominal voltage range but within the operating voltage range, the device will exhibit normal functional behavior. However, parametric specifications may not be specified outside the nominal supply voltage range.

Table 11. Operating Voltage Range

V_{BB} Voltage Range	Note
3 V to 6 V	Transients such as cold crank and start-stop, functional operation are specified but some parametric specifications may not apply. The device is completely short-circuit protected up to 125°C.
6 V to 18 V	Nominal supply voltage, all parametric specifications apply. The device is completely short-circuit protected up to 125°C.
18 V to 40 V	Transients such as jump-start and load-dump, functional operation specified but some parametric specifications may not apply.

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12 Layout

12.1 Layout Guidelines

To achieve optimal thermal performance, connect the exposed pad to a large copper pour. On the top PCB layer, the pour may extend beyond the package dimensions as shown in the example below. In addition to this, it is recommended to have a V_{BB} plane either on one of the internal PCB layers or on the bottom layer.

Vias should connect this plane to the top V_{BB} pour.

Ensure that all external components are placed close to the pins. Device current limiting performance can be harmed if the R_{ILIM} is far from the pins and extra parasitics are introduced.

12.2 Layout Example

The layout example shown below is for device versions A/B/C.

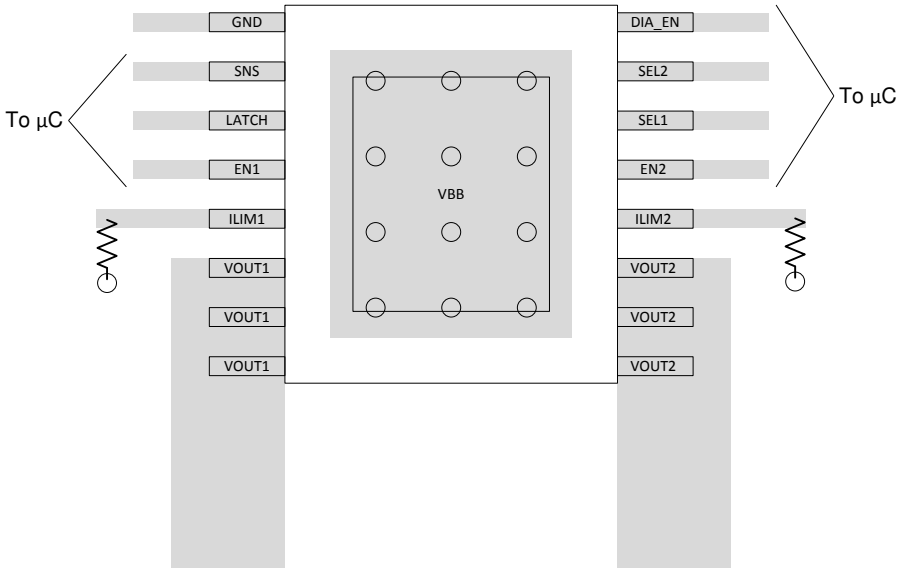


Figure 57. PWP Layout Example

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13 Device and Documentation Support

13.1 Documentation Support

13.1.1 Related Documentation

For related documentation see the following:

- TI's *How To Drive Inductive, Capacitive, and Lighting Loads with Smart High Side Switches*
- TI's *Short Circuit Reliability Test for Smart Power Switch*
- TI's *Adjustable Current Limit of Smart Power Switches*
- TI's *TPS2HB35-Q1 40-V, 35-mΩ Dual-Channel Smart High-Side Switch*
- TI's *Improved Automotive Short Circuit Reliability with Adjustable Current Limiting*

13.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

13.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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13.4 Trademarks

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13.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

13.6 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

14 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TPS2HB35AQPWPRQ1	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	ROHS Exempt	NIPDAU	Level-3-260C-168HRS	-40 to 125	2HB35AQ
TPS2HB35AQPWPRQ1.A	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	ROHS Exempt	NIPDAU	Level-3-260C-168HRS	-40 to 125	2HB35AQ
TPS2HB35AQPWPRQ1.B	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	-	Call TI	Call TI	-40 to 125	
TPS2HB35BQPWPRQ1	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	ROHS Exempt	NIPDAU	Level-3-260C-168HRS	-40 to 125	2HB35BQ
TPS2HB35BQPWPRQ1.A	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	ROHS Exempt	NIPDAU	Level-3-260C-168HRS	-40 to 125	2HB35BQ
TPS2HB35BQPWPRQ1.B	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	-	Call TI	Call TI	-40 to 125	
TPS2HB35CQPWPRQ1	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	ROHS Exempt	NIPDAU	Level-3-260C-168HRS	-40 to 125	2HB35CQ
TPS2HB35CQPWPRQ1.A	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	ROHS Exempt	NIPDAU	Level-3-260C-168HRS	-40 to 125	2HB35CQ
TPS2HB35CQPWPRQ1.B	Active	Production	HTSSOP (PWP) | 16	3000 | LARGE T&R	-	Call TI	Call TI	-40 to 125	

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

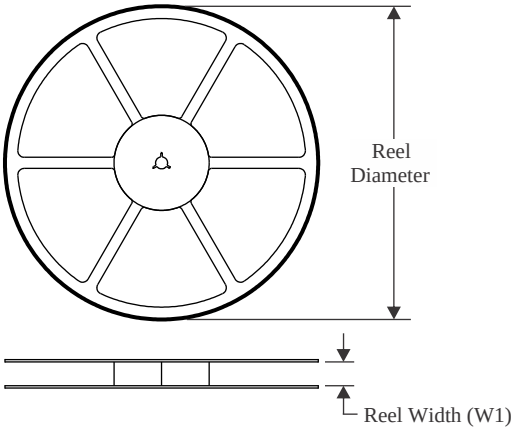
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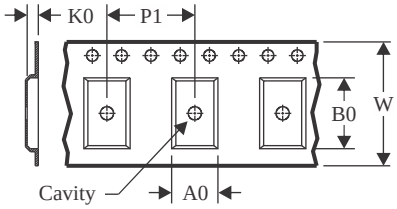
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION

REEL DIMENSIONS

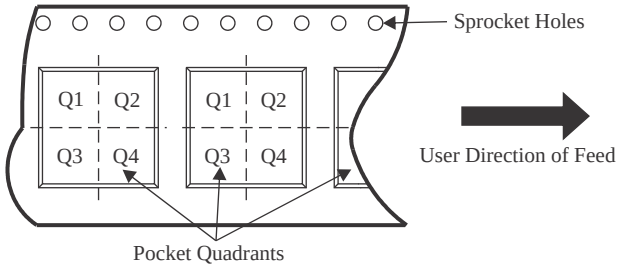


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

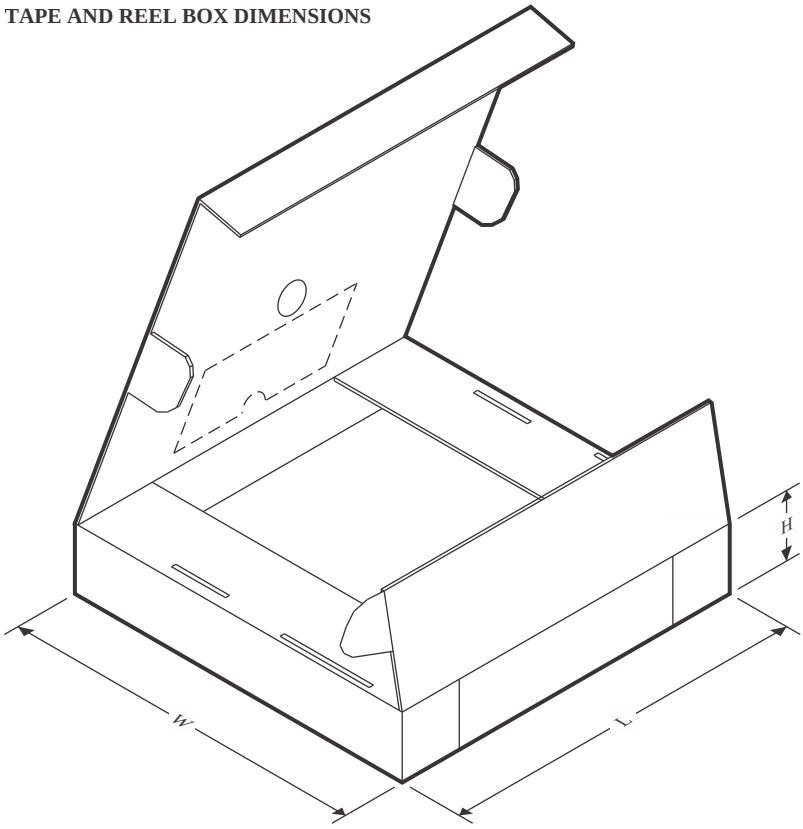
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TPS2HB35AQPWPRQ1	HTSSOP	PWP	16	3000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TPS2HB35BQPWPRQ1	HTSSOP	PWP	16	3000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TPS2HB35CQPWPRQ1	HTSSOP	PWP	16	3000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS

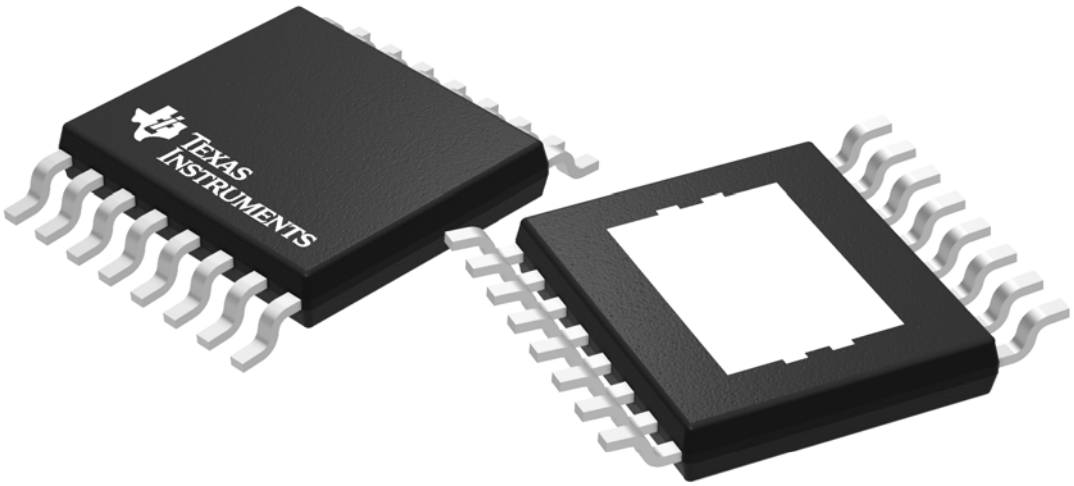


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TPS2HB35AQPWPRQ1	HTSSOP	PWP	16	3000	350.0	350.0	43.0
TPS2HB35BQPWPRQ1	HTSSOP	PWP	16	3000	350.0	350.0	43.0
TPS2HB35CQPWPRQ1	HTSSOP	PWP	16	3000	350.0	350.0	43.0

PWP 16 PowerPAD™ TSSOP - 1.2 mm max height

PLASTIC SMALL OUTLINE



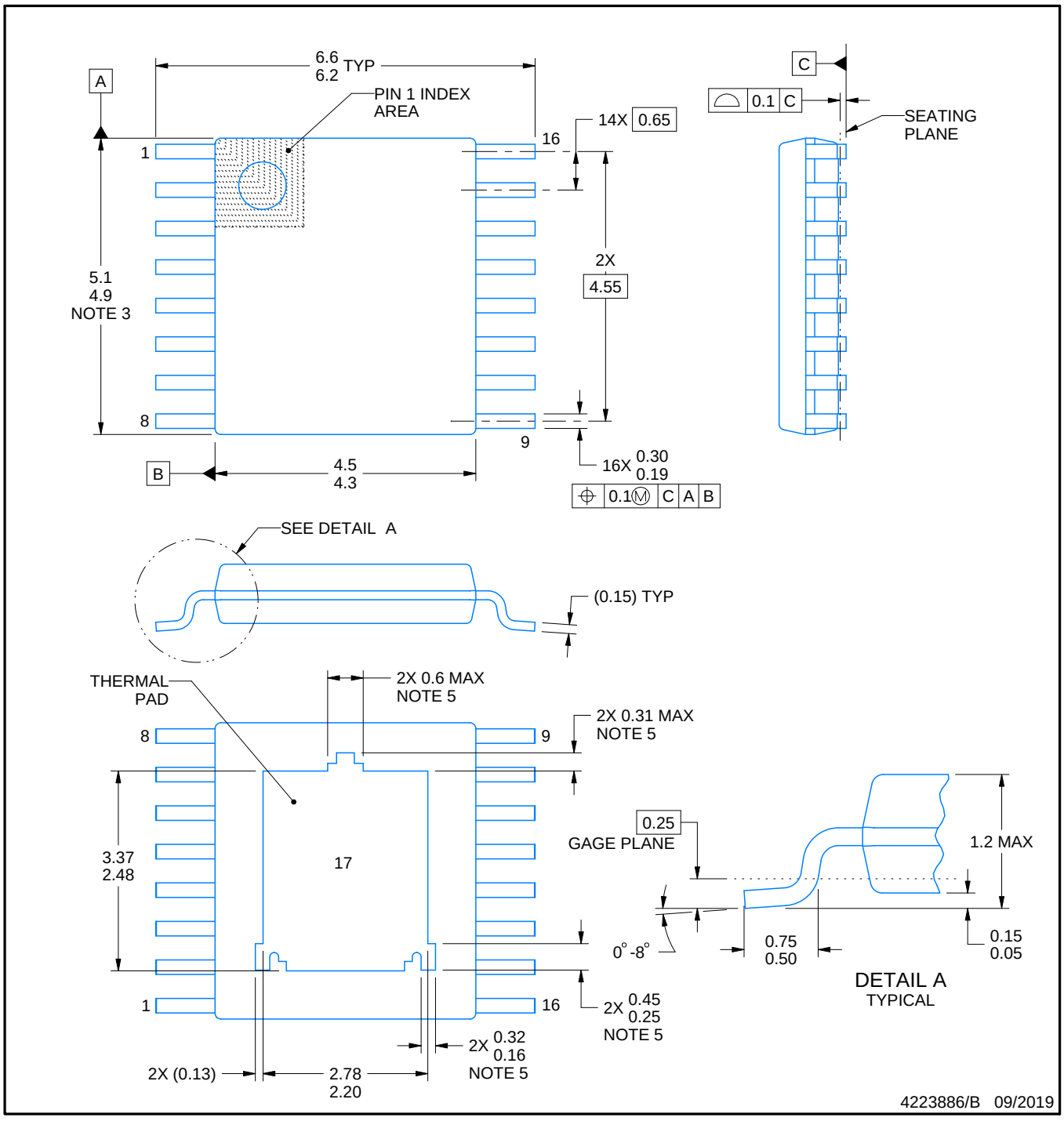
Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

PACKAGE OUTLINE

PWP0016M

PowerPAD™ TSSOP - 1.2 mm max height

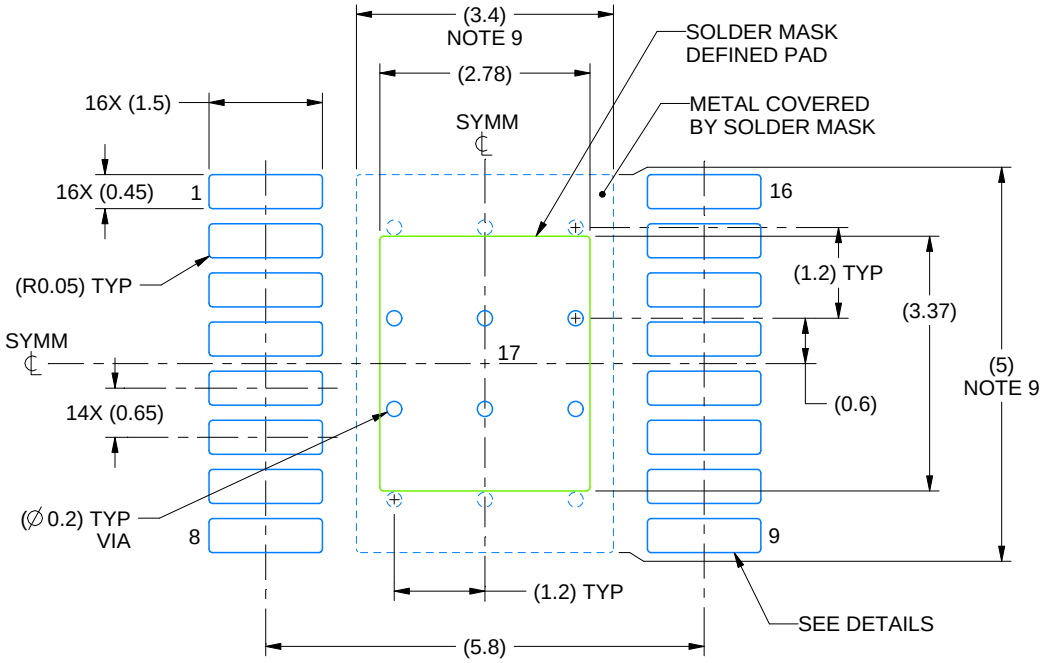
SMALL OUTLINE PACKAGE



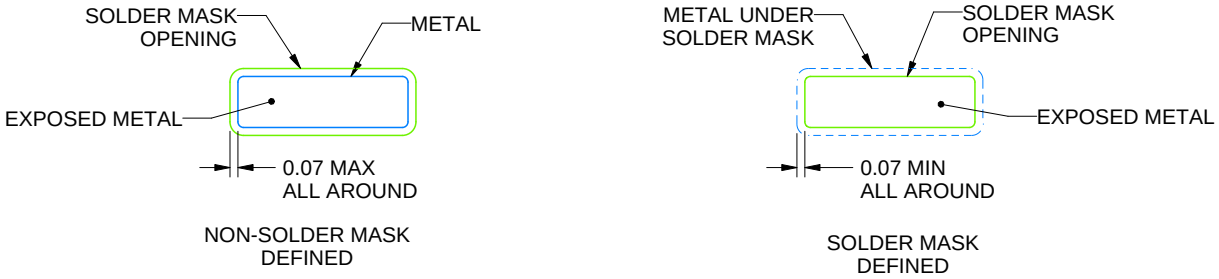
PowerPAD is a trademark of Texas Instruments.

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-153.
5. Features may differ or may not be present.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X

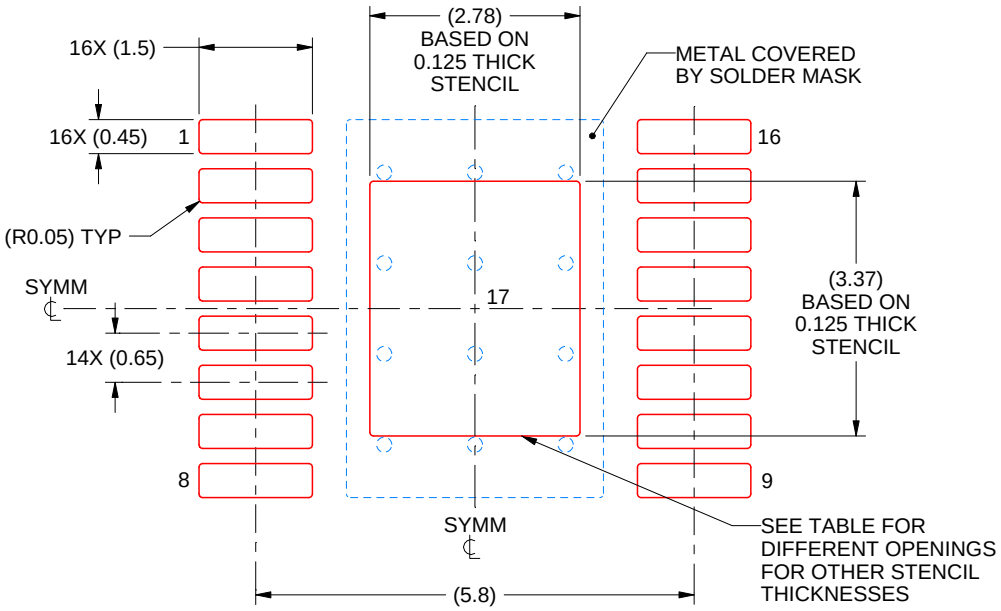


SOLDER MASK DETAILS

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature numbers SLMA002 (www.ti.com/lit/slma002) and SLMA004 (www.ti.com/lit/slma004).
9. Size of metal pad may vary due to creepage requirement.
10. Vias are optional depending on application, refer to device data sheet. It is recommended that vias under paste be filled, plugged or tented.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

STENCIL THICKNESS	SOLDER STENCIL OPENING
0.1	3.11 X 3.77
0.125	2.78 X 3.37 (SHOWN)
0.15	2.54 X 3.08
0.175	2.35 X 2.85

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NOTES: (continued)

11. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
12. Board assembly site may have different recommendations for stencil design.

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